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The equation $x^4 + xy + 4y^3 + y + 1 = 0$ has no integer solutions

Size $h = 55$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + 4y^3 + y + 1 = 0 \tag{1}$$

has no integer solutions.

Replacing y by $-y$ transforms (1) into $f(x, y) = 0$, where

$$f(x, y) = 4y^3 + xy + y - x^4 - 1, \tag{2}$$

that is, $4y^3 + xy + y = x^4 + 1$. We prove that $f(x, y) = 0$ has no integer solutions; all statements below refer to the coordinates of (2).

The proof combines two reciprocity arguments. In Section 2, a graph of cubic Hilbert symbols over $\mathbb{Q}(\omega)$, with 76 linear auxiliary polynomials and 542 edges, shows that every integer solution satisfies

$$(x, y) \bmod 7 \in \{(0, 4), (0, 5), (2, 2), (4, 6)\}. \tag{3}$$

In Section 3, a graph of Hilbert symbols of order 8 over $\mathbb{Q}(\zeta_8)$, with 77 linear auxiliary polynomials and 519 edges, defines an element $\mathcal{A}$ of the Brauer group of the curve $f = 0$. At an integer solution the local invariants of $\mathcal{A}$ would have to add up to 0; in Section 4 we show that they cannot on the residue classes (3).

Section 2 uses only the explicit formula for tame Hilbert symbols and the Hilbert reciprocity law. Section 3 also quotes standard facts on Brauer groups of curves: residues and their compatibility with corestriction, purity, and the description of the local evaluation maps by the Tate pairing. Apart from these, the proof uses polynomial identities modulo the restrictions of f to the auxiliary lines and finite computations with residues; all their data are listed in Section 6. The whole proof is also formalized in Lean 4 in the file `Solution.lean`, see Section 5.

1 Hilbert symbols of order 3 and 8

The fields. Let ω be a primitive cube root of unity and ζ a primitive 8th root of unity, so that $\omega^2 + \omega + 1 = 0$ and $\zeta^4 = -1$. For $n = 3$ we put $K_3 = \mathbb{Q}(\omega)$ and $\zeta_3 = \omega$, and for $n = 8$ we put $K_8 = \mathbb{Q}(\zeta)$ and $\zeta_8 = \zeta$. Both fields are totally complex; their rings of integers are $\mathbb{Z}[\omega]$ and $\mathbb{Z}[\zeta]$. In K_3 we write $\langle A, B\rangle = A + B\omega$, also when A and B are polynomials in x and y; then $\langle A, B\rangle\langle C, D\rangle = \langle AC - BD,\ AD + BC - BD\rangle$, and the norm is $N(\langle A, B\rangle) = A^2 - AB + B^2$. The Galois group of $K_8/\mathbb{Q}$ is $G = \{\sigma_1, \sigma_3, \sigma_5, \sigma_7\}$, where $\sigma_c(\zeta) = \zeta^c$; we identify G with $(\mathbb{Z}/8\mathbb{Z})^\times$, and note that $c^2 \equiv 1 \pmod 8$ for every $c \in G$.

For a finite place v of K_n we write $K_{n,v}$ for the completion, $v(a)$ for the normalized valuation, and call a a unit at v if $v(a) = 0$. In K_3, a prime $p \equiv 2 \pmod 3$ is inert, with residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod 3$, the polynomial $t^2 + t + 1$ has two roots r modulo p, and for each of them there is a place v_r above p, at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p; its residue field is $\mathbb{F}_p$, and ω reduces to r. The prime 3 is ramified: $\pi = 1 - \omega$ generates the prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$. In K_8, an odd prime p has four places above it with residue field $\mathbb{F}_p$ if $p \equiv 1 \pmod 8$, and two places with residue field $\mathbb{F}_{p^2}$ otherwise; 2 is totally ramified, with uniformizer $\lambda = 1 - \zeta$.

Hilbert symbols. For a place v of K_n and $a, b \in K_{n,v}^\times$, the Hilbert symbol $(a,b)_v$ of order n is an nth root of unity [1, Chapter V, §3]. We write $(a,b)_v = \zeta_n^{\beta_v(a,b)}$ with $\beta_v(a,b) \in \mathbb{Z}/n\mathbb{Z}$, and we use the following facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a,bc) = \beta_v(a,b) + \beta_v(a,c)$, $\beta_v(a,b) = -\beta_v(b,a)$, and $\beta_v(a,c^n) = 0$.

(H2) Let v be a finite place not above n, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v. Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w, and define $\chi_v(u) \in \mathbb{Z}/n\mathbb{Z}$ by $\bar{u}^{(q-1)/n} = \bar{\zeta}_n^{\chi_v(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a,b) = e\,\chi_v(w) - f\,\chi_v(u) + ef\,\chi_v(-1)$. In particular, $\beta_v(a,b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K_n^\times$, $\beta_v(a,b) = 0$ for all but finitely many places v, and $\sum_v \beta_v(a,b) = 0$, where the sum is over all places of K_n. At the complex places, $\beta_v(a,b) = 0$.

For $n = 3$, $\chi_v(-1) = 0$ because $-1 = (-1)^3$. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same. For $n = 8$ we also use the behaviour under G: the symbol is compatible with field isomorphisms, $(\sigma a, \sigma b)_{\sigma v} = \sigma\big((a,b)_v\big)$, and $\sigma_c(\zeta^\beta) = \zeta^{c\beta}$, hence

$$\beta_{\sigma_c v}(\sigma_c a, \sigma_c b) = c\,\beta_v(a,b). \tag{4}$$

Classes in K_3 at the places not above 3. Let $p \neq 3$ be a prime and v a place of K_3 above p. For $a = p^e u \in K_{3,v}^\times$ with a unit u, put $c_v(a) = (e \bmod 3,\ \chi_v(u)) \in \mathbb{F}_3^2$. By (H1) and (H2), c_v is additive, $c_v(a)$ determines the class of a modulo cubes, and $\beta_v(a,b) = e\chi_v(w) - f\chi_v(u)$ for $c_v(a) = (e, \chi_v(u))$ and $c_v(b) = (f, \chi_v(w))$. We put $c_p(a) = c_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod 3$, and $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod 3$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, β_p is the sum of the forms β_v over the places v above p. For $p \equiv 2 \pmod 3$ and a rational integer u prime to p, we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi_v(u) = 0$.

The place above 3 in K_3.

Lemma 2. *Let v be the place of K_3 above 3. Every non-zero element of $K_{3,v}$ is equal to $\pi^a \omega^b 2^c (1+3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $c_3(A) = (a,b,c,d) \in \mathbb{F}_3^4$. Then c_3 is additive, and $\beta_v(A,B) = c_3(A)^{\mathsf{T}} M_3\, c_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod 3.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod \pi$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1+3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in $K_{3,v}$, and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1+3\omega)^d$ with $b, c, d \in \{0,1,2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements π, ω, 2 and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), β_v at the place above 3 is minus the sum

of the values at the place above 2 and at the two places above 7. These places correspond to $\bar\omega = 2$ and $\bar\omega = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar\omega = 2$	$\bar\omega = 4$
π	$(0,2)$	$(0,0)$	$(0,2)$
ω	$(0,1)$	$(0,2)$	$(0,2)$
2	$(1,0)$	$(0,2)$	$(0,1)$
$1+3\omega$	$(0,2)$	$(1,0)$	$(0,0)$

where, at the places above 7, $\chi(u)$ is defined by $\bar u^2 = \bar\omega^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3. □

We write β_3 for the form of Lemma 2. Since $3 = -\omega^2\pi^2$ and $4 = 2^2$, we have $c_3(3) = (2,2,0,0)$ and $c_3(4) = (0,0,2,0)$.

2 The cubic graph

The data. The cubic graph consists of 76 polynomials $H_i = \langle P_i, Q_i\rangle$, $0 \le i \le 75$, where $P_i, Q_i \in \mathbb{Z}[x,y]$ have degree at most 1, of constants $c_i \in \mathbb{Z}[\omega]$, and of 542 edges (i,j) with $i<j$ and weights $e_{ij} \in \{1,2\}$. They are listed in Section 6.1. The vertices $0,\dots,5$ are *real*: $Q_i = 0$. The other 70 vertices form 35 pairs of complex conjugate vertices $i, \bar\imath$, with $H_{\bar\imath} = \bar H_i$, and the graph is invariant under complex conjugation: with an edge (i,j) of weight e it contains the edge $(\bar\imath, \bar\jmath)$ of weight $-e$, after reordering its endpoints (an edge between i and $\bar\imath$ is its own conjugate).

Let (x,y) be an integer solution of $f(x,y) = 0$ at which all the values $H_i = H_i(x,y)$ are non-zero. For a place v of K_3 put

$$B_v = \sum_{(i,j)} e_{ij}\,\beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \tag{5}$$

the first sum over the edges, and $B_p = \sum_{v|p} B_v$ for every prime p.

Lemma 3. *At such a solution,* $\sum_p B_p = 0$.

Proof. Apply (H3) to every pair (H_i, H_j) of an edge and to every pair (H_i, c_i), multiply by the weights and add; the complex place contributes nothing. □

Lemma 4. *No H_i vanishes at an integer solution of $f(x,y) = 0$.*

Proof. For integers x, y, $H_i(x,y) = 0$ if and only if $P_i(x,y) = Q_i(x,y) = 0$, since 1 and ω are linearly independent over $\mathbb{Q}$. For a real vertex this is a line defined over $\mathbb{Q}$, and for the other vertices it is a point with rational coordinates. A direct computation shows that the only rational point of $f = 0$ on these lines and points is $(0, \frac12)$, which lies on $H_0, H_5, H_{34}, H_{35}, H_{36}, H_{37}$, and it is not integral. □

Good primes. Let i be a vertex, and write $H_i = \alpha_i x + \beta_i y + \gamma_i$ with $\alpha_i, \beta_i, \gamma_i \in \mathbb{Z}[\omega]$. If $\beta_i \ne 0$, we parametrize the line $H_i = 0$ by $t = x$, $y = -(\alpha_i t + \gamma_i)/\beta_i$, put $a_i = \beta_i$ and $f_i(t) = \beta_i^3 f\big(t, y(t)\big)$; otherwise by $x = -\gamma_i/\alpha_i$, $t = y$, and put $a_i = \alpha_i$ and $f_i(t) = \alpha_i^4 f\big(x,t\big)$. Then $f_i \in \mathbb{Z}[\omega][t]$, with leading coefficient $-\beta_i^3$, respectively $4\alpha_i^4$. Let $g_{ij}(t) \in K_3[t]$ be the restriction of H_j to the line, so that $a_i g_{ij} \in \mathbb{Z}[\omega][t]$. Put $\rho_{ij} = e_{ij}$ and $\rho_{ji} = 3 - e_{ij}$ for an edge (i,j), $i<j$, so that $\rho_{ij} \in \{1,2\}$, $\rho_{ij} = 0$ if i and j are not joined, and

$$A_i(t) = c_i \prod_j g_{ij}(t)^{\rho_{ij}}.$$

We use two kinds of identities. For every edge (i,j) with $i<j$,

$$\alpha_{ij}(t)f_i(t)+\gamma_{ij}(t)\,a_i g_{ij}(t)=D_{ij} \tag{6}$$

with polynomials α_{ij},γ_{ij} over $\mathbb{Z}[\omega]$ and a non-zero constant $D_{ij}\in\mathbb{Z}[\omega]$; we take $D_{ij}=\mathrm{Res}_t(f_i,a_ig_{ij})$. For every vertex i, with an integer $D_i\geq 1$ and a polynomial $w_i\in\mathbb{Z}[\omega][t]$,

$$D_i^3A_i(t)\equiv w_i(t)^3 \pmod{f_i(t)} \quad \text{in } K_3[t]. \tag{7}$$

Proposition 5. *Suppose that* (6) *holds for every edge and* (7) *for every vertex. Let S be a finite set of primes containing* 2 *and* 3 *and the primes dividing the norms of the c_i, of the a_i, of the D_{ij}, and the D_i. Then $B_p=0$ for every prime $p\notin S$, at every integer solution.*

Proof. Let v be a place above $p\notin S$. By (H2), every term of (5) in which both arguments are units at v vanishes. Let i be a vertex with $v(H_i)>0$, and let t be the value of the parameter at the solution. Since a_i is a unit at v, the other coordinate given by the parametrization is congruent modulo v to that of the solution; hence $f_i(t)\equiv 0$ and $g_{ij}(t)\equiv H_j$ modulo v, and all the coefficients involved are v-integral. If v also divides H_j for an edge (i,j), (6) (for the smaller of i,j) gives $v\mid D_{ij}$, which is impossible. Hence every edge with a non-zero term has exactly one endpoint divisible by v, and by (H1) all terms involving a vertex i with $v\mid H_i$ combine to $\beta_v(H_i, c_i\prod_j H_j^{\rho_{ij}})$, where we used $\rho_{ji}\equiv -e_{ij} \pmod 3$. The quotient of $D_i^3A_i-w_i^3$ by f_i has coefficients whose denominators divide powers of 2 and of a_i, because the leading coefficient of f_i is $-a_i^3$ or $4a_i^4$; so (7) gives $D_i^3c_i\prod_j H_j^{\rho_{ij}}\equiv w_i(t)^3$ modulo v, and the left side is a unit at v. Hence $c_i\prod_j H_j^{\rho_{ij}}$ is a non-zero cube in the residue field, and the symbol vanishes by (H2). Hence $B_v=0$, and $B_p=0$. □

Lemma 6. *The hypotheses of Proposition 5 hold with*

$$S=\{2,3,5,7,13,19,37\}.$$

Proof. The norms of the constants c_i are $\{2,3,7,19\}$-units, and the norms of the a_i are $\{2,3\}$-units. For every edge, Table 2 gives the factorization of the norm of D_{ij}; all are supported on $\{2,3,5,7,13,19\}$. For every vertex, Table 1 gives D_i, which is a $\{2,3,37\}$-unit, and w_i; then (7) holds, as division by f_i shows. □

The graph is the closure under complex conjugation of a graph with 276 edges in which the vertex identities only hold in a relaxed form, $A_i\equiv \bar{A}_{\bar{\imath}}W_i^3$; taking the closure turns them into the identities (7) and multiplies all local values by 2.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 7. *Let p be a prime, $k\geq 1$, and let $z\in\mathbb{Z}[\omega]$ be non-zero with $z\equiv\zeta \pmod{p^k}$, where $\zeta=\langle A,B\rangle$ with $0\leq A,B<p^k$. Then $c_p(z)\in b+D$ for the vector b and the subspace D given as follows.*

(i) $p\equiv 2 \pmod 3$. *If $\zeta\neq 0$, then $b=c_p(\zeta)$ and $D=0$. If $\zeta=0$, then $b=0$, and D is $\mathbb{F}_3^2$, or the span of $(1,0)$ if $z\in\mathbb{Z}$.*

(ii) $p\equiv 1 \pmod 3$. *For each root r, let ρ be the root of t^2+t+1 modulo p^k congruent to r. If $A+B\rho\not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.*

(iii) $p=3$. *If $\zeta\neq 0$, let $a=v(\zeta)<2k$ be its valuation at π and $m=2k-a$. Then $b=c_3(\zeta)$, and $D=0$ if $m\geq 4$, while $D=D_m$ if $m\leq 3$, where D_1 is spanned by $(0,1,0,0)$, $(0,0,1,0)$ and $(0,0,0,1)$, D_2 by $(0,0,1,0)$ and $(0,0,0,1)$, and D_3 by $(0,0,1,1)$. If $\zeta=0$, then $b=0$ and $D=\mathbb{F}_3^4$.*

(iv) $p = 3$, $z \in \mathbb{Z}$, *so that* $B = 0$. *If* $3^k \nmid A$, *let* $j = v_3(A)$. *Then* $b = c_3(A)$ *and* $D = 0$ *if* $k - j \geq 2$, *while* $b = j\, c_3(3)$ *and* D *is spanned by* $c_3(4)$ *if* $k - j = 1$. *If* $3^k \mid A$, *then* $b = 0$ *and* D *is spanned by* $c_3(3)$ *and* $c_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod p$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r, z and ζ are the p-adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m. (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. □

Lemma 8. *Let* p *be a prime, and consider vertices* i *with vectors* b_i, *subspaces* D_i *and vectors* γ_i, *and a set of edges* (i, j) *with* $i < j$ *and weights* $e_{ij} \in \{1, 2\}$. *With* ρ_{ij} *as above, put* $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. *Suppose that* $\beta_p(d, \sigma_i) = 0$ *for every* i *and every* $d \in D_i$, *and that* $\beta_p(d, d') = 0$ *for every edge* (i, j) *and all* $d \in D_i$, $d' \in D_j$. *Then*

$$\sum_{(i,j)} e_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} e_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ *for every* i. *By bilinearity, it suffices to check the hypotheses for* d *and* d' *in spanning sets of* D_i *and* D_j.

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} e_{ij} \beta_p(d_i, d_j)$, and both sums vanish. □

Local values. Let $p \in S$. Here 2 and 5 are inert, the places above 7 are v_2 and v_4, those above 13 are v_3 and v_9, those above 19 are v_7 and v_{11}, and those above 37 are v_{10} and v_{26}. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i(x, y) \equiv H_i(a, b) \pmod{p^k}$, and Lemma 7, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for the real vertices, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 8 hold, then B_p is determined by a, b and k. Starting from the solutions of $f \equiv 0 \pmod p$, we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of $f \equiv 0 \pmod{p^{k+1}}$, until these hypotheses hold, or until no solution remains. Table 3 lists the resulting 1068 terminal classes and the value of B_p in each; every point of $U(\mathbb{Z}_p)$, where U is the curve $f = 0$, lies in one of them.

Lemma 9. *At every integer solution of* $f(x, y) = 0$,

$$B_2 = 1, \quad B_3 = 2, \quad B_5 = 0, \quad B_{13} = 0, \quad B_{19} = 2, \quad B_{37} = 0,$$

and $B_7 = 1$ *if* $(x, y) \bmod 7$ *is one of* $(0,4), (0,5), (2,2), (4,6)$, *while* $B_7 = 2$ *if it is one of* $(1,3), (3,2), (5,1)$. *These seven residue classes are all the solutions of* $f \equiv 0 \pmod 7$, *and all of them are smooth points modulo* 7.

Proof. This is read off from Table 3: for $p \neq 7$ all the classes have the value stated, and for $p = 7$ the value in each class is the one stated for its reduction modulo 7. □

Proposition 10. *Every integer solution of* $f(x, y) = 0$ *satisfies* (3).

Proof. By Lemma 4, the sums B_p are defined, and by Lemma 3, Proposition 5 and Lemma 6, $B_2 + B_3 + B_5 + B_7 + B_{13} + B_{19} + B_{37} = 0$ in $\mathbb{Z}/3\mathbb{Z}$. By Lemma 9 this reads $1 + 2 + 0 + B_7 + 0 + 2 + 0 = 0$, that is, $B_7 = 1$. □

3 The graph of order 8

The curve. Let U be the affine curve $f(x,y)=0$ over $\mathbb{Q}$, and X its closure in $\mathbb{P}^2$. The curve X is a smooth plane quartic, of genus 3, and its only point at infinity is the rational point $\infty=[0:1:0]$. Write $K=K_8=\mathbb{Q}(\zeta)$. The curve U contains the rational point $Q_0=(0,\frac12)$ and the four points $R_k=(\zeta^k,0)$, $k\in\{1,3,5,7\}$, which form one closed point of U with residue field K. We call these five points the *special points*.

Brauer classes. For a field F of characteristic 0 containing ζ and $a,b\in F^\times$ let $(a,b)_8\in\mathrm{Br}(F)$ be the class of the cyclic algebra defined by ζ, a and b; it is bilinear, $(a,b)_8=-(b,a)_8$, and $8(a,b)_8=0$. For a completion of a number field containing ζ, $\mathrm{inv}(a,b)_8=\beta(a,b)/8$ with the β of Section 1 [1, Chapter V, §3]. Applying $\sigma_c\in G$ to the coefficients maps $(a,b)_8$ to $c\,(\sigma_c a,\sigma_c b)_8$, by the same argument as (4). We use the following standard facts on the Brauer group of the smooth projective curve X [2]:

(B1) Every class $\alpha\in\mathrm{Br}(\mathbb{Q}(X))$ has a residue $\partial_x\alpha\in H^1(\mathbb{Q}(x),\mathbb{Q}/\mathbb{Z})$ at every closed point x of X; $\alpha\in\mathrm{Br}(X)$ if and only if all these residues vanish (purity), and then α can be evaluated at every point of X. For a finite extension $L/\mathbb{Q}$, residues commute with corestriction: $\partial_x\,\mathrm{Cor}_{L(X)/\mathbb{Q}(X)}\,\beta=\sum_{y\mapsto x}\mathrm{Cor}_{L(y)/\mathbb{Q}(x)}\,\partial_y\beta$.

(B2) The corestrictions to $\mathbb{Q}$ of the residues of α at all closed points add up to 0.

(B3) For a regular proper model $\mathcal{X}$ over $\mathbb{Z}_p$, a class in $\mathrm{Br}(X\otimes\mathbb{Q}_p)$ whose residue along the special fibre vanishes extends to $\mathrm{Br}(\mathcal{X})$, and then $\mathrm{inv}_p\,\alpha(P)=0$ for all $P\in X(\mathbb{Q}_p)$.

The data. Let $g_1,\dots,g_{25}\in\mathbb{Z}[\zeta]$ be the elements listed in Section 6.2: $g_1=\zeta$ and $g_2=1+\zeta+\zeta^2$ are units, $g_3=1+\zeta$ lies above 2, and the others are generators of prime ideals above $3,5,7,11,13,17,41,73$. Put

$$S=\{2,3,5,7,11,13,17,41,73\}.$$

The graph of order 8 consists of 77 lines $H_i=\alpha_i x+\beta_i y+\gamma_i$ with $\alpha_i,\beta_i,\gamma_i\in\mathbb{Z}[\zeta]$, of constants c_i, which are products of the g_k, and of 519 edges (i,j) with $i<j$ and weights $e_{ij}\in\{1,\dots,7\}$; see Section 6.2. Put

$$B=\sum_{(i,j)}e_{ij}(H_i,H_j)_8+\sum_i(H_i,c_i)_8\in\mathrm{Br}(K(X)),\qquad \mathcal{A}=\mathrm{Cor}_{K(X)/\mathbb{Q}(X)}\,B\in\mathrm{Br}(\mathbb{Q}(X)).$$

For a prime p and a point $P\in U(\mathbb{Q}_p)$ at which all $H_i(P)$ are non-zero, let

$$b_p(P)=\sum_{v\mid p}\Big(\sum_{(i,j)}e_{ij}\beta_v\big(H_i(P),H_j(P)\big)+\sum_i\beta_v\big(H_i(P),c_i\big)\Big)\in\mathbb{Z}/8\mathbb{Z},\tag{8}$$

the sum over the places v of K above p. Then $\mathrm{inv}_p\,\mathcal{A}(P)=b_p(P)/8$, because the invariant of a corestriction is the sum of the invariants.

Lemma 11. *No H_i vanishes at an integer solution of $f(x,y)=0$.*

Proof. For integers x,y, $H_i(x,y)=0$ means four linear equations with integer coefficients, the coefficients of $1,\zeta,\zeta^2,\zeta^3$. For 56 lines they have no rational solution. For 15 lines their only rational solution is one point, which is $(0,\frac12)$ or one of $(1,1)$, $(0,0)$, $(-1,-1)$, $(-1,1)$, $(1,0)$, $(-1,0)$, where f takes the non-zero values $4,-1,-6,2,-2,-2$. The remaining six lines are defined over $\mathbb{Q}$: $H_{26}=y$, $H_{27}=2y-1$, $H_{30}=x+1$, $H_{40}=x$, $H_{49}=x+2y-1$ and $H_{70}=x-1$. On them f restricts to $-x^4-1$, to a function of the non-integer $y=\frac12$, to $4y^3-2$, to $(2y-1)(2y^2+y+1)$, to $-2(2y-1)(4y^3-7y^2+3y-1)$ with $x=1-2y$, and to $2(2y^3+y-1)$, none of which has an integer zero. □

The Galois closure. The 77 lines have 152 distinct Galois conjugates $\sigma_c(H_i)$, where σ_c acts on the coefficients, and they form 49 orbits under G. We regard B as a graph on these 152 lines, the 75 lines not listed carrying neither edges nor constants. By the rule $(a,b)_8 \mapsto c\,(\sigma_c a, \sigma_c b)_8$ and $\mathrm{Cor}\circ\sigma_c^* = \mathrm{Cor}$, moving an edge or a constant from a line to a conjugate line, with the weight multiplied by c, does not change $\mathcal{A}$; the graph listed in Section 6.2 was obtained in this way from a graph whose edges are placed at representatives of the orbits.

For a line ℓ of the closure put $A_\ell = c_\ell \prod_j H_j^{\rho_{\ell j}}$, where $\rho_{ij} = e_{ij}$ and $\rho_{ji} = -e_{ij}$ for an edge (i,j), and $c_\ell = 1$, $A_\ell = 1$ for the lines not listed; A_ℓ is a function on $\ell = 0$. Let $\Gamma_\ell \subseteq G$ be the stabilizer of ℓ, and put

$$\tilde{A}_\ell = \prod_c \sigma_c\big(A_{\sigma_c\ell}\big)^c, \qquad N_\Gamma(A) = \prod_{m\in\Gamma} \sigma_m(A)^m,$$

the first product over representatives c of G/Γ_ℓ, where $\sigma_c\ell$ is the line $\sigma_c(\ell)$ and σ_c acts on the coefficients; $\tilde{A}_\ell$ is a function on the line $\ell = 0$, and $N_{\Gamma_\ell}(\tilde{A}_\ell) = \prod_{c\in G} \sigma_c(A_{\sigma_c\ell})^c$ does not change under the moves just described. For a field $F \supseteq K$ let $\kappa\colon F^\times \to H^1(F, \mathbb{Z}/8\mathbb{Z})$ be the Kummer map with respect to ζ. The edges of B join lines which meet at an *anchor*, a point $P \in K^2$ at which $f(P)$ is a non-zero S-unit; in particular two lines joined by an edge never meet on X.

Proposition 12. *(i) Let x be a closed point of U, not a special point, lying on the line ℓ and on no line of another G-orbit, and suppose that no point of X_K above x lies on two different lines of the orbit of ℓ. Let y be a point of X_K above x on $\ell = 0$, μ the intersection multiplicity of ℓ and X at y, and $D \subseteq \Gamma_\ell$ the image of $\mathrm{Gal}(K(y)/\mathbb{Q}(x))$ in G. Then $\partial_x\mathcal{A} = \mu\,\mathrm{Cor}_{K(y)/\mathbb{Q}(x)}\,\kappa(Z_y)$ with $Z_y = \prod_\gamma \sigma_\gamma(\tilde{A}_\ell)(y)^\gamma$, the product over representatives γ of Γ_ℓ/D, and its restriction to $K(y)$ is $\mu\kappa\big(N_{\Gamma_\ell}(\tilde{A}_\ell)(y)\big)$. If $N_{\Gamma_\ell}(\tilde{A}_\ell)(y)$ is an 8th power in $K(y)$, then $\partial_x\mathcal{A}$ is a homomorphism $D \to 4\mathbb{Z}/8\mathbb{Z}$, whose value at the Frobenius of a prime $\mathfrak{p} \nmid 2$ of $\mathbb{Q}(x)$ of degree one, unramified in $K(y)$, at which Z_y is a unit, is $\mu\sum_{\mathfrak{P}|\mathfrak{p}} \chi_{\mathfrak{P}}(Z_y)$.*

(ii) At the closed point $\{R_k\}$, $\partial\mathcal{A} = \kappa(Z_R)$, where we identify its residue field with K by R_1 and

$$Z_R = \prod_{m\in G} \prod_{\ell\ni R_m} \sigma_m\big(A_\ell(R_m)\big)^{m\mu_\ell(R_m)},$$

the inner product over the lines ℓ of the closure through R_m, with the intersection multiplicity $\mu_\ell(R_m)$ of ℓ and X at R_m. At Q_0, $\partial\mathcal{A} = \mathrm{Cor}_{K/\mathbb{Q}}\,\kappa(Z_Q)$ with $Z_Q = \prod_{\ell\ni Q_0} A_\ell(Q_0)^{\mu_\ell(Q_0)}$; it vanishes if and only if $N_G(Z_Q)$ is an 8th power in K and $\sum_{w|q}\chi_w(Z_Q) = 0$ for one prime $q \equiv 3$ and one prime $q \equiv 5 \pmod 8$ at which Z_Q is a unit.

If x lies on lines of several orbits, its residue is the sum of the contributions of the orbits, since lines through a common point of X are never joined by an edge.

Proof. By (B1), $\partial_x\mathcal{A} = \sum_{y'} \mathrm{Cor}_{K(y')/\mathbb{Q}(x)}\,\partial_{y'}B$, the sum over the points y' of X_K above x, and $\partial_{y'}B = \mu\kappa(A_j(y'))$ if y' lies on the line j only, because the lines joined to j by edges do not vanish at y'. The points above x are the $\sigma_c(y)$, $c \in G/D$, and $\sigma_c(y)$ lies only on the line $\sigma_c(\ell)$ of the orbit; in particular $D \subseteq \Gamma_\ell$. The automorphism σ_c of X_K over X maps y to $\sigma_c(y)$ and acts on K by σ_c; since the Kummer map for ζ^c is c^{-1} times the Kummer map for ζ and $c^{-1} \equiv c \pmod 8$, transporting the residue at $\sigma_c(y)$ to y gives $\mu\kappa\big(\sigma_c(A_{\sigma_c\ell})(y)^c\big)$, without changing the corestriction. Writing $c = c'\gamma$ with c' in G/Γ_ℓ and γ in Γ_ℓ/D and using $\sigma_{c'\gamma}(\ell) = \sigma_{c'}(\ell)$, the sum of these terms is $\mu\kappa(Z_y)$. The restriction of $\mathrm{Cor}_{K(y)/\mathbb{Q}(x)}$ to $K(y)$ is $\sum_{\delta\in D}\delta^*$, and δ fixes the coordinates of y, which lie in $\mathbb{Q}(x)$; so the restriction is $\mu\kappa(N_{\Gamma_\ell}(\tilde{A}_\ell)(y))$. If this vanishes, the residue lies in the kernel of the restriction, which is $\mathrm{Hom}(D, \mathbb{Z}/8\mathbb{Z})$, of exponent 2 since $D \subseteq (\mathbb{Z}/8\mathbb{Z})^\times$; the value of a corestriction at $\mathrm{Frob}_\mathfrak{p}$ is the sum of the values at the $\mathrm{Frob}_\mathfrak{P}$, $\mathfrak{P} \mid \mathfrak{p}$ [1]. Part (ii) is the same computation with all lines through the point: above $\{R_k\}$ lie the four points R_m, with residue field K, and the residue at R_m, $\kappa\big(\prod_{\ell\ni R_m} A_\ell(R_m)^{\mu_\ell(R_m)}\big)$, is transported to $R_1 = \sigma_m(R_m)$ by σ_m; above Q_0 lies the single point Q_0, with residue field K. Finally, $\mathrm{Cor}_{K/\mathbb{Q}}\,\kappa(Z)$ restricts to

$\kappa(N_G(Z))$ on K, and the kernel of the restriction, $\mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z})$, is detected by the Frobenius elements σ_3 and σ_5. □

The identities on the lines. Table 4 lists a representative ℓ of each of the 49 orbits, as $\sigma_c(H_j)$ for a listed line H_j. If $\ell = \alpha x + \beta y + \gamma$, we parametrize $\ell = 0$ by $t = x$, $y = -(\alpha t + \gamma)/\beta$ if $\beta \neq 0$, and by $x = -\gamma/\alpha$, $t = y$ otherwise. Let $f_\ell(t)$ be the restriction of f to the line, divided by its leading coefficient, and ψ_ℓ the quotient of f_ℓ by $(t - t_P)^{\mu_P}$ for the special points P on $\ell = 0$, where t_P is the parameter of P and μ_P its multiplicity as a root of f_ℓ; the table lists these points. Write $N_{\Gamma_\ell}(\tilde{A}_\ell) = \kappa_\ell \prod_m m^{n_m}$, the product over the lines m of the closure, with $\kappa_\ell = \prod_{c \in G} \sigma_c(c_{\sigma_c \ell})^c \in K^\times$ and $n_m \in \mathbb{Z}$, and put

$$P_\ell(t) = \kappa_\ell \prod_m m(t)^{\bar{n}_m}, \qquad \bar{n}_m \in \{0, \dots, 7\},\ \bar{n}_m \equiv n_m \pmod 8,$$

where $m(t)$ is the restriction of m to $\ell = 0$. Since no line m with $n_m \neq 0$ meets $\ell = 0$ on X (it is a conjugate of a line joined by an edge to a conjugate of ℓ), $N_{\Gamma_\ell}(\tilde{A}_\ell)(y)$ and $P_\ell(y)$ differ by a non-zero 8th power at every point y of X_K on $\ell = 0$.

Proposition 13. *The class $\mathcal{A}$ lies in* $\mathrm{Br}(X)$.

Proof. The following finite computations with exact arithmetic verify the hypotheses of Proposition 12 at every closed point of U.

- The 519 edges, at 481 anchors, join lines meeting at points P with S-integral coordinates at which $f(P)$ is a non-zero S-unit; the lines are primitive outside S, and the constants are S-units.
- For the 193 pairs of distinct lines of one orbit, the lines meet on X only at special points.
- For the representative ℓ of each orbit (Table 4), ψ_ℓ is squarefree, and if $\deg \psi_\ell > 0$, which is the case for 48 of the 49 orbits, then $P_\ell \equiv W_\ell^8 \pmod{\psi_\ell}$ in $K[t]$ for the polynomial W_ℓ of the table. Hence $N_{\Gamma_\ell}(\tilde{A}_\ell)(y)$ is an 8th power in $K(y)$ at every point y of X_K on $\ell = 0$ other than the special points; by the transport in the proof of Proposition 12, the same holds for every line of the orbit.
- For the 18 representatives with non-trivial stabilizer and $\deg \psi_\ell > 0$, Table 5 lists the 19 irreducible factors q of ψ_ℓ over K and the subgroup D of the $c \in \Gamma_\ell$ with $\sigma_c(q) = q$. For each generator σ_m of D (σ_3 and σ_5 if $D = G$), it lists a prime $p \equiv m \pmod 8$ and a simple root $t_0 \in \mathbb{F}_p$ of q modulo a place of K above p at which the coefficients of q reduce to $\mathbb{F}_p$, and the point (x_0, y_0) of the line modulo p with parameter t_0. This defines a prime $\mathfrak{p}$ of degree one of $\mathbb{Q}(x)$ with Frobenius σ_m in D, and the value of the homomorphism of Proposition 12(i) at it is 0; 24 checks in all.
- Z_R and $N_G(Z_Q)$ are 8th powers in K, see (9), and $\sum_{w|q} \chi_w(Z_Q) = 0$ for $q = 307 \equiv 3$ and $q = 317 \equiv 5 \pmod 8$.

Hence all residues of $\mathcal{A}$ at the closed points of U vanish. The only other closed point of X is the rational point ∞, so by (B2) the residue at ∞ vanishes as well, and $\mathcal{A} \in \mathrm{Br}(X)$ by (B1). □

The values of the lines at Q_0 and R_1 that do not vanish there are S-units, and

$$\begin{aligned} Z_Q &= g_1^{7} g_2^{-74} g_3^{190} g_4^{9} g_5^{27} g_6^{13} g_7^{21} g_8^{2} g_9^{13} g_{10}^{6} g_{11}^{16} g_{14}^{8} g_{15}^{10} g_{16}^{9} g_{17}^{23} g_{18}^{6} g_{19}^{3} g_{20}^{2} g_{21}^{3} g_{22}^{15} g_{23}^{3} g_{24}^{11} g_{25}^{7}, \\ N_G(Z_Q) &= \big(g_2^{-190} g_3^{380} g_4^{45} g_5^{27} g_6^{36} g_7^{32} g_8^{15} g_9^{15} g_{10}^{27} g_{11}^{17} g_{14}^{27} g_{15}^{30} g_{16}^{23} g_{17}^{20} g_{18}^{6} g_{19}^{8} g_{20}^{7} g_{21}^{7} g_{22}^{13} g_{23}^{23} g_{24}^{15} g_{25}^{21}\big)^8, \\ Z_R &= \big(g_2^{-462} g_3^{786} g_4^{93} g_5^{63} g_6^{16} g_7^{12} g_8^{23} g_9^{12} g_{12}^{5} g_{13}^{3} g_{14}^{13} g_{15}^{47} g_{16}^{39} g_{17}^{20} g_{18}^{6} g_{19}^{10} g_{20}^{9} g_{21}^{7} g_{22}^{5} g_{23}^{9} g_{24}^{7} g_{25}^{7}\big)^8. \end{aligned} \tag{9}$$

Proposition 14. *For every prime $p \notin S$ and every $P \in U(\mathbb{Z}_p)$,* $\mathrm{inv}_p\, \mathcal{A}(P) = 0$.

Proof. The primes of bad reduction of the model $f = 0$ are 2, 3, 277, 1321 and 167149: on the singular points, $f_y = 0$ gives $x = -1 - 12y^2$, and the resultant with respect to y of f_x and f after this substitution is $2^{41} \cdot 3^{16} \cdot 277 \cdot 1321 \cdot 167149$; the point at infinity is smooth over $\mathbb{Z}_p$ for odd p. At each of these primes there is exactly one singular point modulo p, namely $(1, 0)$, $(2, 2)$, $(237, 225)$, $(686, 936)$ and $(17882, 157066)$, and $v_p(f) = 1$ at every lift of it, so that no point of $U(\mathbb{Z}_p)$ reduces to it; at the odd ones the special fibre is irreducible with a node there, at which the model is regular. Let $p \notin S$. Then p is odd, the closure of U in $\mathbb{P}^2_{\mathbb{Z}_p}$ is regular with irreducible special fibre, all the data of B are p-integral, the lines are primitive and the constants are units at the places $w \mid p$ of K. So no H_i vanishes identically on the special fibre, the residue of B along the special fibre over the ring of integers of K_w vanishes for every $w \mid p$, and since $K_w/\mathbb{Q}_p$ is unramified, the residue of $\mathcal{A}$ along the special fibre, the sum of the corestrictions of these, vanishes as well. By (B3), $\operatorname{inv}_p \mathcal{A} = 0$ on $U(\mathbb{Z}_p)$. □

Local values at the primes of S. Let J be the Jacobian of X and $\iota\colon X \to J$, $P \mapsto [P - \infty]$. Since $\mathcal{A} \in \operatorname{Br}(X)$ and $8\mathcal{A} = 0$, there are $\xi \in H^1(\mathbb{Q}, J)[8]$ and constants γ_v such that $\operatorname{inv}_v \mathcal{A}(P) = \langle \iota(P), \xi_v \rangle_v + \gamma_v$ for every place v and every $P \in X(\mathbb{Q}_v)$, where $\langle\ ,\ \rangle_v$ is the local Tate pairing [3, 4]. At an odd prime $p \in S$, $\operatorname{inv}_p \mathcal{A}$ is therefore constant on the residue disc of each smooth point of the special fibre, since the kernel of reduction of the Néron model is pro-p, hence divisible by 8. At 2, $U(\mathbb{Z}_2)$ is the union of the residue discs of $(0, 1)$ and $(1, 1)$, on which x, respectively y, is a parameter of the smooth locus of the model; two points of such a disc whose parameters agree modulo 2^5 have images in the Néron model whose difference lies in $\hat{J}(2^5\mathbb{Z}_2) = 8\hat{J}(4\mathbb{Z}_2) \subseteq 8J(\mathbb{Q}_2)$, because the logarithm of the formal group is an isomorphism onto $(2^k\mathbb{Z}_2)^3$ for $k \geq 2$; so $\operatorname{inv}_2 \mathcal{A}$ is constant on the 32 discs modulo 2^5.

We computed b_p from (8) at two points of each of the 168 smooth residue discs at the odd primes of S (1, 5, 7, 11, 9, 22, 35 and 78 discs for $p = 3, 5, 7, 11, 13, 17, 41, 73$), and at the 32 discs modulo 2^5 at 2, with 2-adic precision 2^{64} and 2^{80}; the results are in Tables 6 and 7. At the places above odd primes we use (H2). At the place λ above 2, every non-zero element a of K_λ is $\lambda^v \zeta^s u_1^{c_1} u_2^{c_2} u_3^{c_3} u_4^{c_4}$ times an 8th power, with the elements

$$u_1 = -2 + 2\zeta + \zeta^3, \quad u_2 = -1 - \zeta + \zeta^2, \quad u_3 = -2 + \zeta^2 + 2\zeta^3, \quad u_4 = -1 + 2\zeta^3$$

of $\mathbb{Z}[\zeta]$, of norms 73, 9, 9 and 17, which are units at λ, for a unique vector $c_2(a) = (v, s, c_1, \ldots, c_4) \in (\mathbb{Z}/8\mathbb{Z})^6$: here v is the valuation, the 2-adic logarithms of the u_s^8 span $\log\big((1 + \lambda\mathbb{Z}_2[\zeta])^8\big)$, so that $(c_1, \ldots, c_4)$ is obtained from the coordinates of $\log(a'^8)$ in this basis, where $a' = a\lambda^{-v}$, and s from the residue of $a' \prod_s u_s^{-c_s}$ modulo 4. Then $\beta_\lambda(a, b) = c_2(a)^{\mathsf{T}} M_2\, c_2(b)$ with

$$M_2 = \begin{pmatrix} 0 & 0 & 3 & 3 & 2 & 3 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 5 & 7 & 4 & 3 & 5 & 4 \\ 5 & 7 & 5 & 4 & 6 & 5 \\ 6 & 7 & 3 & 2 & 4 & 2 \\ 5 & 6 & 4 & 3 & 6 & 0 \end{pmatrix} \pmod 8,$$

the symbols of the basis elements being obtained from (H2) at the places above 3, 17 and 73 and (H3). The results are as follows.

Lemma 15. *For every $P = (x, y) \in U(\mathbb{Z}_p)$:*

p	$8\,\mathrm{inv}_p\,\mathcal{A}(P)$
2	0 *if* x *is odd,* 2 *if* $x \equiv 2 \pmod 4$, 6 *if* $x \equiv 0 \pmod 4$
5	2
7	$6, 1, 7, 5, 6, 6, 4$ *for* $(x, y) \bmod 7 = (0,4), (0,5), (1,3), (2,2), (3,2), (4,6), (5,1)$
17	4
$3, 11, 13, 41, 73$	0

Proof. This is read off from Tables 6 and 7; the residue discs at 7 are those of the seven points modulo 7. □

4 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ satisfies $f(x, y) = 0$, and let $P = (x, y)$. By Proposition 10, $(x, y) \bmod 7$ is one of $(0,4), (0,5), (2,2), (4,6)$, and by Lemma 15, $8\,\mathrm{inv}_7\,\mathcal{A}(P) \in \{6, 1, 5\}$. By Proposition 13, $\mathcal{A}(P) \in \mathrm{Br}(\mathbb{Q})$ is defined, and $\sum_v \mathrm{inv}_v\,\mathcal{A}(P) = 0$, the sum over all places of $\mathbb{Q}$. The real place contributes 0, because $\mathcal{A}$ is a corestriction from the totally complex field K. (Alternatively, by Lemma 11 all $H_i(P)$ are non-zero, and the relation $\sum_p b_p(P) = 0$ follows directly from (H3) applied to (8).) By Proposition 14 and Lemma 15,

$$0 = \sum_p 8\,\mathrm{inv}_p\,\mathcal{A}(P) = d + 2 + s + 4 \in \mathbb{Z}/8\mathbb{Z}, \qquad d \in \{0, 2, 6\},\ s \in \{1, 5, 6\}.$$

But $d + s + 6$ takes the values $7, 3, 4$ for $d = 0$, $1, 5, 6$ for $d = 2$, and $5, 1, 2$ for $d = 6$, never 0. This contradiction proves Theorem 1. □

5 Verification and formalization

Computations. All identities, residue conditions and local values above were computed with exact arithmetic. The data of Section 6 were then checked again by a separate program that reads them from the source of this file: the vertex identities and resultants of the cubic graph, the residue classes of Table 3 with Lemmas 7 and 8 and the covering of $U(\mathbb{Z}_p)$, the 8th power identities of Table 4 on the Galois closure of the listed graph, the checks of Table 5, the elements (9) with the values at 307 and 317, the matrix M_2, and the data of Proposition 14. An independent re-evaluation of b_p at the 336 odd sample points gave the values of Table 6.

Lean. The file `Solution.lean` proves Theorem 1 in Lean 4 with Mathlib; its proof uses only the axioms `propext`, `Quot.sound` and `Classical.choice`, and `comparator` checks that it proves exactly the statement of `Challenge.lean`. The formal proof follows the structure above, with two differences. It proves the reciprocity laws for the cubic and the 8th power Hilbert symbols itself, within a development of local and global class field theory contained in the file. And it avoids purity and the Tate pairing: the vanishing outside S is derived directly from the 8th power identities of the graph on its Galois closure, separately at points reducing to a special point and at the other points, and at the primes of S the values are computed on explicit finite covers of the integral points by residue classes (5, 53, 25, 21, 9, 1750, 675 and 582 classes for $p = 3, 5, 7, 11, 13, 17, 41, 73$, and 512 classes modulo 2^9 at 2). The file is large (about 58 MB), and checking it takes about 12 hours.

6 The data

6.1 The cubic graph

The vertices $H_i = \langle P_i, Q_i\rangle = P_i + Q_i\omega$, with r for the real vertices and the conjugate vertex $\bar\imath$ otherwise, and the constants $c_i = \langle a, b\rangle = a + b\omega$:

i	H_i		c_i	i	H_i		c_i
0	$\langle -x+2y-1,\ 0\rangle$	r	$\langle -21, 84\rangle$	38	$\langle y-1,\ -x+1\rangle$	$\bar\imath = 39$	$\langle 3, 3\rangle$
1	$\langle x-1,\ 0\rangle$	r	$\langle 40071, 11913\rangle$	39	$\langle x+y-2,\ x-1\rangle$	$\bar\imath = 38$	$\langle -9, -9\rangle$
2	$\langle x+y+1,\ 0\rangle$	r	$\langle 189, 567\rangle$	40	$\langle x+y-1,\ x+1\rangle$	$\bar\imath = 41$	$\langle 1, 1\rangle$
3	$\langle y+1,\ 0\rangle$	r	$\langle 0, -27\rangle$	41	$\langle y-2,\ -x-1\rangle$	$\bar\imath = 40$	$\langle -1, -1\rangle$
4	$\langle x+1,\ 0\rangle$	r	$\langle 0, -27\rangle$	42	$\langle x-1,\ 2\rangle$	$\bar\imath = 43$	$\langle -9, -9\rangle$
5	$\langle x,\ 0\rangle$	r	$\langle -14, 7\rangle$	43	$\langle x-3,\ -2\rangle$	$\bar\imath = 42$	$\langle 3, 3\rangle$
6	$\langle -x+y-1,\ -x-1\rangle$	$\bar\imath = 7$	$\langle -156, 64\rangle$	44	$\langle -x+2y-1,\ x+3\rangle$	$\bar\imath = 45$	$\langle -8, -12\rangle$
7	$\langle y,\ x+1\rangle$	$\bar\imath = 6$	$\langle 6, 16\rangle$	45	$\langle -2x+2y-4,\ -x-3\rangle$	$\bar\imath = 44$	$\langle -128, -48\rangle$
8	$\langle x+y-1,\ -1\rangle$	$\bar\imath = 9$	$\langle -1, 1\rangle$	46	$\langle x-2,\ -3\rangle$	$\bar\imath = 47$	$\langle 144, -351\rangle$
9	$\langle x+y,\ 1\rangle$	$\bar\imath = 8$	$\langle 3, 3\rangle$	47	$\langle x+1,\ 3\rangle$	$\bar\imath = 46$	$\langle 24, 9\rangle$
10	$\langle x+y-1,\ x-1\rangle$	$\bar\imath = 11$	$\langle -20, 12\rangle$	48	$\langle -x+y,\ -x-1\rangle$	$\bar\imath = 49$	$\langle -3, 0\rangle$
11	$\langle y,\ -x+1\rangle$	$\bar\imath = 10$	$\langle 880, 624\rangle$	49	$\langle y+1,\ x+1\rangle$	$\bar\imath = 48$	$\langle 1, 2\rangle$
12	$\langle y-1,\ -1\rangle$	$\bar\imath = 13$	$\langle 0, 1\rangle$	50	$\langle x+y,\ -1\rangle$	$\bar\imath = 51$	$\langle 9, 0\rangle$
13	$\langle y,\ 1\rangle$	$\bar\imath = 12$	$\langle 0, 1\rangle$	51	$\langle x+y+1,\ 1\rangle$	$\bar\imath = 50$	$\langle -3, 0\rangle$
14	$\langle x+2y-1,\ x-1\rangle$	$\bar\imath = 15$	$\langle -8, -3\rangle$	52	$\langle y,\ -1\rangle$	$\bar\imath = 53$	$\langle -3, 0\rangle$
15	$\langle 2y,\ -x+1\rangle$	$\bar\imath = 14$	$\langle -2, -3\rangle$	53	$\langle y+1,\ 1\rangle$	$\bar\imath = 52$	$\langle 9, 0\rangle$
16	$\langle x-1,\ -1\rangle$	$\bar\imath = 17$	$\langle -2, 2\rangle$	54	$\langle 2y,\ -1\rangle$	$\bar\imath = 55$	$\langle -4, 4\rangle$
17	$\langle x,\ 1\rangle$	$\bar\imath = 16$	$\langle 12, 12\rangle$	55	$\langle 2y+1,\ 1\rangle$	$\bar\imath = 54$	$\langle 48, 48\rangle$
18	$\langle y-1,\ -x-2\rangle$	$\bar\imath = 19$	$\langle 147, 147\rangle$	56	$\langle x,\ -1\rangle$	$\bar\imath = 57$	$\langle -2, 2\rangle$
19	$\langle x+y+1,\ x+2\rangle$	$\bar\imath = 18$	$\langle -7, 7\rangle$	57	$\langle x+1,\ 1\rangle$	$\bar\imath = 56$	$\langle 12, 12\rangle$
20	$\langle y-1,\ -2\rangle$	$\bar\imath = 21$	$\langle 144, 0\rangle$	58	$\langle y,\ -2\rangle$	$\bar\imath = 59$	$\langle 0, 12\rangle$
21	$\langle y+1,\ 2\rangle$	$\bar\imath = 20$	$\langle -12, 0\rangle$	59	$\langle y+2,\ 2\rangle$	$\bar\imath = 58$	$\langle 0, 144\rangle$
22	$\langle 2y-1,\ x-2\rangle$	$\bar\imath = 23$	$\langle -16, -55\rangle$	60	$\langle -x+2y,\ -2\rangle$	$\bar\imath = 61$	$\langle 196, 196\rangle$
23	$\langle -x+2y+1,\ -x+2\rangle$	$\bar\imath = 22$	$\langle 8, 5\rangle$	61	$\langle -x+2y+2,\ 2\rangle$	$\bar\imath = 60$	$\langle -38416, -38416\rangle$
24	$\langle -x+3y-1,\ -2x-2\rangle$	$\bar\imath = 25$	$\langle 7, 14\rangle$	62	$\langle x,\ -2\rangle$	$\bar\imath = 63$	$\langle 32, 4\rangle$
25	$\langle x+3y+1,\ 2x+2\rangle$	$\bar\imath = 24$	$\langle -147, 0\rangle$	63	$\langle x+2,\ 2\rangle$	$\bar\imath = 62$	$\langle 768, -240\rangle$
26	$\langle -x+3y-1,\ x-2\rangle$	$\bar\imath = 27$	$\langle -4, -4\rangle$	64	$\langle -x+2y+1,\ -x-1\rangle$	$\bar\imath = 65$	$\langle -9, -9\rangle$
27	$\langle -2x+3y+1,\ -x+2\rangle$	$\bar\imath = 26$	$\langle 2, 2\rangle$	65	$\langle 2y+2,\ x+1\rangle$	$\bar\imath = 64$	$\langle 3, 3\rangle$
28	$\langle x-1,\ -2\rangle$	$\bar\imath = 29$	$\langle -2, 3\rangle$	66	$\langle 2x+1,\ -2\rangle$	$\bar\imath = 67$	$\langle -44, 4\rangle$
29	$\langle x+1,\ 2\rangle$	$\bar\imath = 28$	$\langle 16, 21\rangle$	67	$\langle 2x+3,\ 2\rangle$	$\bar\imath = 66$	$\langle 2288, 368\rangle$
30	$\langle x-1,\ -3\rangle$	$\bar\imath = 31$	$\langle -57, -38\rangle$	68	$\langle y+1,\ -x\rangle$	$\bar\imath = 69$	$\langle -3, 0\rangle$
31	$\langle x+2,\ 3\rangle$	$\bar\imath = 30$	$\langle -1083, -2888\rangle$	69	$\langle x+y+1,\ x\rangle$	$\bar\imath = 68$	$\langle 1, 2\rangle$
32	$\langle y-1,\ -x\rangle$	$\bar\imath = 33$	$\langle 4, 0\rangle$	70	$\langle 2y+1,\ -x\rangle$	$\bar\imath = 71$	$\langle 2, 2\rangle$
33	$\langle x+y-1,\ x\rangle$	$\bar\imath = 32$	$\langle 16, 0\rangle$	71	$\langle x+2y+1,\ x\rangle$	$\bar\imath = 70$	$\langle -4, -4\rangle$
34	$\langle -x+2y-1,\ -x\rangle$	$\bar\imath = 35$	$\langle -6, -9\rangle$	72	$\langle -x+y,\ -x\rangle$	$\bar\imath = 73$	$\langle 48, -80\rangle$
35	$\langle 2y-1,\ x\rangle$	$\bar\imath = 34$	$\langle -72, -27\rangle$	73	$\langle y,\ x\rangle$	$\bar\imath = 72$	$\langle -12, -4\rangle$
36	$\langle -x+2y-1,\ -2x\rangle$	$\bar\imath = 37$	$\langle 0, 12\rangle$	74	$\langle y,\ -x\rangle$	$\bar\imath = 75$	$\langle 4, 2\rangle$
37	$\langle x+2y-1,\ 2x\rangle$	$\bar\imath = 36$	$\langle 0, 144\rangle$	75	$\langle x+y,\ x\rangle$	$\bar\imath = 74$	$\langle 0, -12\rangle$

The 542 edges: the row of i lists the vertices $j > i$ joined to i, with the weight e_{ij} as exponent.
0: $10^1\ 11^2\ 12^2\ 13^1\ 14^2\ 15^1\ 16^1\ 17^2\ 28^1\ 29^2\ 54^2\ 55^1\ 56^2\ 57^1\ 64^2\ 65^1\ 66^1\ 67^2\ 68^2\ 69^1\ 74^1\ 75^2$ **1**: $22^1\ 23^2\ 32^2\ 33^1\ 34^2\ 35^1\ 44^2\ 45^1\ 54^1\ 55^2\ 60^1\ 61^2\ 64^1\ 65^2\ 68^1\ 69^2$ **2**: $22^1\ 23^2\ 36^2\ 37^1\ 54^2\ 55^1\ 58^2\ 59^1\ 70^2\ 71^1$ **3**: $8^2\ 9^1\ 22^2\ 23^1\ 36^1\ 37^2\ 56^2\ 57^1$ **4**: $14^1\ 15^2\ 54^1\ 55^2$ **5**: $8^2\ 9^1\ 14^2\ 15^1\ 54^2\ 55^1\ 64^1\ 65^2$ **6**: $8^1\ 21^2\ 25^2\ 35^1\ 45^1\ 50^2\ 54^1\ 56^1\ 58^2\ 62^2\ 70^1\ 74^1$ **7**: $9^2\ 20^1\ 24^1\ 34^2\ 44^2\ 51^1\ 55^2\ 57^2\ 59^1\ 63^1\ 71^2\ 75^2$ **8**: $10^2\ 12^2\ 14^1\ 17^2\ 19^1\ 29^1\ 36^1\ 46^2\ 49^2\ 57^2\ 69^1\ 72^2\ 75^1$ **9**: $11^1\ 13^1\ 15^2\ 16^1\ 18^2\ 28^2\ 37^2\ 47^1\ 48^1\ 56^1\ 68^2\ 73^1\ 74^2$ **10**: $25^1\ 34^2\ 35^2\ 42^1\ 44^1\ 46^2\ 54^2\ 58^2\ 59^1\ 68^1$ **11**: $24^2\ 34^1\ 35^1\ 43^2\ 45^2\ 47^1\ 55^1\ 58^2\ 59^1\ 69^2$ **12**: $24^1\ 27^1\ 31^2\ 32^2\ 34^1\ 57^2\ 62^1\ 63^2\ 68^2\ 69^2$ **13**: $25^2\ 26^2\ 30^1\ 33^1\ 35^2\ 56^1\ 62^1\ 63^2\ 68^1\ 69^1$ **14**: $15^2\ 18^2\ 20^1\ 24^1\ 27^1\ 28^2\ 29^1\ 33^2\ 35^2\ 36^1\ 37^2\ 39^2\ 43^1\ 45^1\ 49^2\ 50^1\ 51^2\ 52^1\ 55^1\ 62^2\ 65^2\ 67^2\ 73^2\ 75^1$ **15**: $19^1\ 21^2\ 25^2\ 26^2\ 28^2\ 29^1\ 32^1\ 34^1\ 36^1\ 37^2\ 38^1\ 42^2\ 44^2\ 48^1\ 50^1\ 51^2\ 53^2\ 54^2\ 63^1\ 64^1\ 66^1\ 72^1\ 74^2$ **16**: $18^2\ 21^1\ 22^2\ 24^1\ 35^1\ 37^1\ 40^1\ 64^1\ 70^2$ **17**: $19^1\ 20^2\ 23^1\ 25^2\ 34^2\ 36^2\ 41^2\ 65^2\ 71^1$ **18**: $19^2\ 21^2\ 29^2\ 48^1\ 54^1\ 56^2\ 58^1\ 60^1\ 61^1\ 64^1\ 65^1\ 69^2\ 71^2\ 73^2$ **19**: $20^1\ 28^1\ 49^2\ 55^2\ 57^1\ 59^2\ 60^2\ 61^2\ 64^2\ 65^2\ 68^1\ 70^1\ 72^1$ **20**: $22^1\ 24^1\ 26^1\ 31^1\ 45^1\ 48^1\ 57^1\ 64^1$ **21**: $23^2\ 25^2\ 27^2\ 30^2\ 44^2\ 49^2\ 56^2\ 65^2$ **22**: $25^2\ 29^2\ 34^1\ 40^2\ 41^2\ 49^1\ 57^2\ 66^1\ 71^2\ 73^2\ 74^2\ 75^2$ **23**: $24^1\ 28^1\ 35^2\ 40^1\ 41^1\ 48^2\ 56^1\ 67^2\ 70^1\ 72^1\ 74^1\ 75^1$ **24**: $25^1\ 34^1\ 36^2\ 39^2\ 48^1\ 49^2\ 54^1\ 55^2\ 58^2\ 60^1\ 61^1$ **25**: $35^2\ 37^1\ 38^1\ 48^1\ 49^2\ 54^1\ 55^2\ 59^1\ 60^2\ 61^2$ **26**: $33^1\ 35^2\ 37^2\ 43^1\ 53^1\ 54^2\ 56^1\ 62^2$ **27**: $32^2\ 34^1\ 36^1\ 42^2\ 52^2$

$55^1\ 57^2\ 63^1$ **28**: $35^1\ 44^1\ 52^2\ 54^1\ 58^1\ 59^2\ 60^2\ 71^2$ **29**: $34^2\ 45^2\ 53^1\ 55^2\ 58^1\ 59^2\ 61^1\ 70^1$ **30**: $50^2\ 61^2\ 65^2$ **31**: $51^1\ 60^1\ 64^1$ **32**: $33^2\ 34^2\ 36^1\ 54^1\ 63^1\ 69^1\ 70^2\ 72^1\ 75^1$ **33**: $35^1\ 37^2\ 55^2\ 62^2\ 68^2\ 71^1\ 73^2\ 74^2$ **34**: $45^2\ 46^1\ 53^1\ 54^1\ 55^1\ 57^2\ 59^2\ 60^2\ 63^2\ 66^1\ 71^1$ **35**: $44^1\ 47^2\ 52^2\ 54^2\ 55^2\ 56^1\ 58^1\ 61^1\ 62^1\ 67^2\ 70^2$ **36**: $38^1\ 39^1\ 40^1\ 42^1\ 45^2\ 46^1\ 49^2\ 57^1\ 59^1\ 60^2\ 61^2\ 70^1\ 75^2$ **37**: $38^2\ 39^2\ 41^2\ 43^2\ 44^1\ 47^2\ 48^1\ 56^2\ 58^2\ 60^1\ 61^1\ 71^2\ 74^1$ **38**: $39^2\ 56^2\ 62^2\ 73^1$ **39**: $57^1\ 63^1\ 72^2$ **40**: $50^2\ 63^1\ 65^1\ 70^1$ **41**: $51^1\ 62^2\ 64^2\ 71^2$ **42**: $61^2\ 71^1$ **43**: $60^1\ 70^2$ **44**: $45^1\ 50^1\ 56^1\ 57^2\ 61^2\ 65^2$ **45**: $51^2\ 56^1\ 57^2\ 60^1\ 64^1$ **46**: $49^2\ 59^2\ 60^2\ 70^1$ **47**: $48^1\ 58^1\ 61^1\ 71^2$ **48**: $54^2\ 55^1\ 64^1\ 70^1$ **49**: $54^2\ 55^1\ 65^2\ 71^2$ **50**: $54^2\ 60^2\ 63^2\ 68^2\ 70^2\ 71^1\ 72^1$ **51**: $55^1\ 61^1\ 62^1\ 69^1\ 70^2\ 71^1\ 73^2$ **52**: $56^1\ 60^1\ 64^2\ 69^2$ **53**: $57^2\ 61^2\ 65^1\ 68^1$ **54**: $57^2\ 66^1\ 69^2\ 71^1$ **55**: $56^1\ 67^2\ 68^1\ 70^2$ **56**: $61^2\ 75^2$ **57**: $60^1\ 74^1$ **58**: 74^2 **59**: 75^1 **60**: $62^1\ 63^2\ 66^2\ 67^2\ 71^1\ 72^1\ 74^1$ **61**: $62^1\ 63^2\ 66^1\ 67^1\ 70^2\ 73^2\ 75^2$ **62**: $64^1\ 70^2\ 71^2$ **63**: $65^2\ 70^1\ 71^1$ **64**: $65^2\ 66^1\ 67^1\ 71^2\ 74^2$ **65**: $66^2\ 67^2\ 70^1\ 75^1$ **66**: $70^1\ 71^2$ **67**: $70^1\ 71^2$ **68**: $69^2\ 72^2$ **69**: 73^1 **70**: $71^1\ 72^1\ 74^2\ 75^2$ **71**: $73^2\ 74^1\ 75^1$ **72**: 75^2 **73**: 74^1 **74**: 75^1

Table 1: the vertex identities (7): the parameter t, D_i and $w_i(t)$, with $\langle a, b\rangle = a + b\omega$.

i	t	D_i	$w_i(t)$
0	x	2^{10}	$\langle 48384, 60480\rangle + \langle -39326077, 29073793\rangle x + \langle -50660134, 37262302\rangle x^2 + \langle -32491221, 23883321\rangle x^3$
1	y	1	$\langle 3648, -1140\rangle + \langle -3838, 1786\rangle y + \langle 6194, 8398\rangle y^2$
2	x	1	$\langle 1827, 0\rangle + \langle 3381, 0\rangle x + \langle 1911, 0\rangle x^2 + \langle -525, 0\rangle x^3$
3	x	1	$\langle -9, -18\rangle + \langle 6, 12\rangle x + \langle -12, -24\rangle x^2 + \langle 9, 18\rangle x^3$
4	y	1	$\langle 0, 6\rangle + \langle 6, 6\rangle y + \langle -12, 0\rangle y^2$
5	y	2^2	$\langle 14, -7\rangle + \langle 42, -7\rangle y + \langle 28, 42\rangle y^2$
6	x	1	$\langle 630, -462\rangle + \langle 1826, -626\rangle x + \langle 2112, -90\rangle x^2 + \langle 780, 366\rangle x^3$
7	x	1	$\langle 882, 1008\rangle + \langle 2142, 2310\rangle x + \langle 2152, 2066\rangle x^2 + \langle 522, 384\rangle x^3$
8	x	1	$\langle -32238, -25194\rangle + \langle 75611, 1411\rangle x + \langle -15065, 75860\rangle x^2 + \langle -28688, -41311\rangle x^3$
9	x	1	$\langle -1866, -34095\rangle + \langle -68389, 26479\rangle x + \langle 109684, 65453\rangle x^2 + \langle -25433, -47929\rangle x^3$
10	x	1	$\langle -1398, 192\rangle + \langle 3816, 180\rangle x + \langle -3354, 72\rangle x^2 + \langle 156, -564\rangle x^3$
11	x	1	$\langle 414, -1278\rangle + \langle 108, 4230\rangle x + \langle 594, -4518\rangle x^2 + \langle -864, 1146\rangle x^3$
12	x	1	$\langle -39, -6\rangle + \langle -3, 30\rangle x + \langle 2, -2\rangle x^2 + \langle 2, 4\rangle x^3$
13	x	1	$\langle 27, 18\rangle + \langle 3, -3\rangle x + \langle 18, 30\rangle x^2 + \langle 0, -15\rangle x^3$
14	x	2^{16}	$\langle -131053890, -7004535\rangle + \langle 51291459, -35733762\rangle x + \langle -123426270, -97463673\rangle x^2 + \langle 78644769, -80338398\rangle x^3$
15	x	2^{16}	$\langle 14312235, 13588638\rangle + \langle 156304662, 75335205\rangle x + \langle -97548027, -73176510\rangle x^2 + \langle 58723290, -73948041\rangle x^3$
16	y	2^2	$\langle -102, 447\rangle + \langle 282, 117\rangle y + \langle 306, 774\rangle y^2$
17	y	2	$\langle 459, 243\rangle + \langle -489, -699\rangle y + \langle 258, 426\rangle y^2$
18	x	1	$\langle 17516016, -1179570\rangle + \langle 39434325, 19439889\rangle x + \langle 24065853, 22653855\rangle x^2 + \langle 5172678, 7827876\rangle x^3$
19	x	1	$\langle 134612226, 37804746\rangle + \langle 178439688, -76316499\rangle x + \langle 48038697, -125388459\rangle x^2 + \langle -6354369, -48555276\rangle x^3$
20	x	1	$\langle -1080, -1056\rangle + \langle -612, -252\rangle x + \langle -12, 72\rangle x^2 + \langle 72, 36\rangle x^3$
21	x	1	$\langle -1944, -3168\rangle + \langle 192, -756\rangle x + \langle 468, 576\rangle x^2 + \langle 36, 228\rangle x^3$
22	x	2^{12}	$\langle 639540, -2873412\rangle + \langle -8375283, -1761228\rangle x + \langle 5779989, 5561046\rangle x^2 + \langle -832599, -2293362\rangle x^3$
23	x	2^7	$\langle 16140, 7284\rangle + \langle -8628, 9549\rangle x + \langle -5826, -675\rangle x^2 + \langle 10734, 1065\rangle x^3$
24	x	3^{15}	$\langle -2181132282, -1409757300\rangle + \langle -1857911580, -1719841914\rangle x + \langle -566744850, -6583689504\rangle x^2 + \langle 3891983424, -2159783514\rangle x^3$
25	x	3^{12}	$\langle 48787620, -94013136\rangle + \langle -181423620, -279815886\rangle x + \langle -293884416, -143559108\rangle x^2 + \langle -67126836, -2167326\rangle x^3$
26	x	3^6	$\langle -3332, -280\rangle + \langle -13782, -4794\rangle x + \langle 1764, 1656\rangle x^2 + \langle 334, 10778\rangle x^3$
27	x	3^7	$\langle 56688, 31356\rangle + \langle -4950, 44082\rangle x + \langle -56448, -60840\rangle x^2 + \langle 50694, 55026\rangle x^3$
28	y	2	$\langle 114, -380\rangle + \langle -470, -378\rangle y + \langle -1032, -732\rangle y^2$
29	y	1	$\langle 1330, 1216\rangle + \langle -884, -880\rangle y + \langle -254, -426\rangle y^2$
30	y	1	$\langle -95, -19\rangle + \langle 19, 57\rangle y$
31	y	1	$\langle 76, -19\rangle + \langle 38, 57\rangle y$
32	x	1	$\langle 32, -344\rangle + \langle 1208, 1090\rangle x + \langle -878, 374\rangle x^2 + \langle -290, -484\rangle x^3$
33	x	1	$\langle 5440, 11600\rangle + \langle 23968, -12496\rangle x + \langle -33040, -27404\rangle x^2 + \langle -2312, 10004\rangle x^3$
34	x	2^{12}	$\langle -1064448, -1257984\rangle + \langle 1418535, -442314\rangle x + \langle -180831, -681702\rangle x^2 + \langle 2403207, 1871082\rangle x^3$

i	t	D_i	$w_i(t)$
35	x	2^{11}	$\langle -628992, -532224\rangle + \langle -8677575, -4168449\rangle x + \langle 954111, 7872009\rangle x^2 + \langle 6377241, 804435\rangle x^3$
36	x	2^{12}	$\langle -508032, -169344\rangle + \langle -19234270098, -16456474431\rangle x + \langle 11356286022, -20641614048\rangle x^2 + \langle 20816482464, 19973709891\rangle x^3$
37	x	2^{12}	$\langle -1693440, -2709504\rangle + \langle -2736671292, -11548209339\rangle x + \langle -18475639650, -6039277326\rangle x^2 + \langle 4401047034, 13428714897\rangle x^3$
38	x	1	$\langle 387, 444\rangle + \langle -741, -282\rangle x + \langle 159, -234\rangle x^2 + \langle 99, 96\rangle x^3$
39	x	1	$\langle 21, -21\rangle + \langle -108, -6\rangle x + \langle 105, 63\rangle x^2 + \langle -18, -24\rangle x^3$
40	x	1	$\langle 60, 237\rangle + \langle 396, 123\rangle x + \langle -45, -240\rangle x^2 + \langle -75, -36\rangle x^3$
41	x	1	$\langle -177, -237\rangle + \langle 273, -123\rangle x + \langle 195, 240\rangle x^2 + \langle -39, 36\rangle x^3$
42	y	1	$\langle 102, 120\rangle + \langle 36, 30\rangle y + \langle 6, 0\rangle y^2$
43	y	1	$\langle -102, 18\rangle + \langle -36, -6\rangle y + \langle -6, -6\rangle y^2$
44	x	2^5	$\langle -1990374, -1763445\rangle + \langle -2356585, -3305288\rangle x + \langle -670676, -1920655\rangle x^2 + \langle 124907, -365132\rangle x^3$
45	x	2^5	$\langle -4206891, -1153581\rangle + \langle -5454575, 749084\rangle x + \langle -1968559, 1909585\rangle x^2 + \langle 63433, 912776\rangle x^3$
46	y	2	$\langle -120, -1965\rangle + \langle 90, -453\rangle y + \langle 30, 108\rangle y^2$
47	y	2	$\langle -55, 283\rangle + \langle -101, -37\rangle y + \langle -10, -44\rangle y^2$
48	x	1	$\langle 6087, 45933\rangle + \langle 201513, 117978\rangle x + \langle 81105, -133836\rangle x^2 + \langle -73749, -72411\rangle x^3$
49	x	1	$\langle 657, 375\rangle + \langle 315, 2412\rangle x + \langle -2556, -315\rangle x^2 + \langle -540, -1056\rangle x^3$
50	x	1	$\langle 14031, 2304\rangle + \langle 58941, 60021\rangle x + \langle -8100, 58833\rangle x^2 + \langle -26280, -12492\rangle x^3$
51	x	1	$\langle -738, -639\rangle + \langle -3192, -120\rangle x + \langle 222, 3324\rangle x^2 + \langle 1290, 681\rangle x^3$
52	x	1	$\langle 1, 5\rangle + \langle -1, 4\rangle x + \langle -2, -1\rangle x^3$
53	x	1	$\langle -21, 0\rangle + \langle 0, 12\rangle x + \langle 3, 3\rangle x^2 + \langle -12, -15\rangle x^3$
54	x	2^6	$\langle 29979, 8766\rangle + \langle 36063, 21708\rangle x + \langle 11718, 27774\rangle x^2 + \langle -22977, 4239\rangle x^3$
55	x	2^6	$\langle -1548, -4311\rangle + \langle -28233, 657\rangle x + \langle -46944, 8532\rangle x^2 + \langle -27630, 16308\rangle x^3$
56	y	2^3	$\langle -882, -813\rangle + \langle -496, -761\rangle y + \langle 830, 4162\rangle y^2$
57	y	2^3	$\langle 314, 109\rangle + \langle -1012, 97\rangle y + \langle 854, 1474\rangle y^2$
58	x	1	$\langle 678, 1194\rangle + \langle -120, -414\rangle x + \langle 90, 126\rangle x^2 + \langle -60, -90\rangle x^3$
59	x	1	$\langle 3744, 324\rangle + \langle -1068, 348\rangle x + \langle 432, 108\rangle x^2 + \langle -300, -60\rangle x^3$
60	x	2^8	$\langle 15789480, -273546\rangle + \langle -35189385, -36824697\rangle x + \langle -15459675, 26509665\rangle x^2 + \langle 20607531, 19429473\rangle x^3$
61	x	2^9	$\langle 147935802, -66101196\rangle + \langle -95649813, 338568489\rangle x + \langle -147429387, -251174637\rangle x^2 + \langle 51765609, 24295425\rangle x^3$
62	y	2	$\langle 2247, -1536\rangle + \langle 1425, -1938\rangle y + \langle 3012, 3540\rangle y^2$
63	y	2^2	$\langle 33048, -11097\rangle + \langle -14166, -10557\rangle y + \langle -4860, 5922\rangle y^2$
64	x	$2^9 \cdot 37$	$\langle -28052073, -32448654\rangle + \langle 10941057, 51599259\rangle x + \langle 40379355, -2239506\rangle x^2 + \langle -23099643, -16057935\rangle x^3$
65	x	2^5	$\langle -2583, -1764\rangle + \langle 9888, -6261\rangle x + \langle 6753, 9384\rangle x^2 + \langle -7158, -3579\rangle x^3$
66	y	2^4	$\langle 3507, -4998\rangle + \langle 12540, 456\rangle y + \langle 11568, 11136\rangle y^2$
67	y	2^2	$\langle 61, -2134\rangle + \langle -2488, -1892\rangle y + \langle -2816, 1184\rangle y^2$
68	x	1	$\langle -14220, 4698\rangle + \langle 6279, 43608\rangle x + \langle 42252, 39642\rangle x^2 + \langle 17493, 1812\rangle x^3$
69	x	1	$\langle -462, -1824\rangle + \langle 2898, -1563\rangle x + \langle 4523, 2626\rangle x^2 + \langle 1013, 1843\rangle x^3$
70	x	2^{12}	$\langle -142186368, -8293332\rangle + \langle -89872500, 172685535\rangle x + \langle 122956164, 264321645\rangle x^2 + \langle 148909896, 105554427\rangle x^3$
71	x	2^{12}	$\langle 10011876, 2515728\rangle + \langle 9269445, 17046600\rangle x + \langle -7588929, 18054840\rangle x^2 + \langle -13953879, 3310812\rangle x^3$
72	x	1	$\langle 1916, 400\rangle + \langle -2008, -2020\rangle x + \langle -1108, -1476\rangle x^2 + \langle 7912, 1896\rangle x^3$
73	x	1	$\langle -42, -10\rangle + \langle 28, -22\rangle x + \langle 18, -28\rangle x^2 + \langle -168, -32\rangle x^3$
74	x	1	$\langle 32946, 27594\rangle + \langle 36360, 1296\rangle x + \langle 36504, -5868\rangle x^2 + \langle -122862, -112734\rangle x^3$
75	x	1	$\langle 5136, -9258\rangle + \langle 15936, 2946\rangle x + \langle 18678, 6162\rangle x^2 + \langle -15594, 39138\rangle x^3$

Table 2: for each edge (i,j), $i<j$, the factorization of the norm of $D_{ij}=\mathrm{Res}_t(f_i,a_ig_{ij})$; the row of i lists j and this factorization.

0: 10: $2^8\cdot 3\cdot 7\cdot 13^2$; 11: $2^8\cdot 3\cdot 7\cdot 13^2$; 12: $2^8\cdot 3\cdot 19$; 13: $2^8\cdot 3\cdot 19$; 14: $2^{16}\cdot 7^2$; 15: $2^{16}\cdot 7^2$; 16: $2^{12}\cdot 7\cdot 13$; 17: $2^{12}\cdot 7\cdot 13$; 28: $2^{16}\cdot 3\cdot 19$; 29: $2^{16}\cdot 3\cdot 19$; 54: $2^{12}\cdot 3\cdot 7\cdot 13$; 55: $2^{12}\cdot 3\cdot 7\cdot 13$; 56: $2^{12}\cdot 7$; 57: $2^{12}\cdot 7$; 64: $2^{16}\cdot 7^2\cdot 13$; 65: $2^{16}\cdot 7^2\cdot 13$; 66: $2^{16}\cdot 5^2\cdot 7$; 67: $2^{16}\cdot 5^2\cdot 7$; 68: $2^8\cdot 3^2\cdot 13$; 69: $2^8\cdot 3^2\cdot 13$; 74: $2^8\cdot 5^2\cdot 13$; 75: $2^8\cdot 5^2\cdot 13$
1: 22: $2^4\cdot 19$; 23: $2^4\cdot 19$; 32: $2^2\cdot 7$; 33: $2^2\cdot 7$; 34: $2^4\cdot 7^2$; 35: $2^4\cdot 7^2$; 44: $2^{10}\cdot 19^2$; 45: $2^{10}\cdot 19^2$; 54: $2^4\cdot 19$; 55: $2^4\cdot 19$; 60: $2^4\cdot 19$; 61: $2^4\cdot 19$; 64: 2^8; 65: 2^8; 68: $2^2\cdot 7\cdot 19$; 69: $2^2\cdot 7\cdot 19$
2: 22: $7^4\cdot 19$; 23: $7^4\cdot 19$; 36: $2^2\cdot 3^2\cdot 5^2\cdot 19$; 37: $2^2\cdot 3^2\cdot 5^2\cdot 19$; 54: $3\cdot 7$; 55: $3\cdot 7$; 58: $2^2\cdot 3\cdot 7^2$; 59: $2^2\cdot 3\cdot 7^2$; 70: $5^2\cdot 7$; 71: $5^2\cdot 7$
3: 8: $2^2\cdot 3\cdot 7$; 9: $2^2\cdot 3\cdot 7$; 22: $2^4\cdot 3^2\cdot 19$; 23: $2^4\cdot 3^2\cdot 19$; 36: $2^2\cdot 3^5\cdot 19$; 37: $2^2\cdot 3^5\cdot 19$; 56: $2^2\cdot 7$; 57: $2^2\cdot 7$
4: 14: $2^8\cdot 3^2$; 15: $2^8\cdot 3^2$; 54: $2^4\cdot 3^2$; 55: $2^4\cdot 3^2$
5: 8: $3\cdot 7$; 9: $3\cdot 7$; 14: $2^4\cdot 7$; 15: $2^4\cdot 7$; 54: $2^4\cdot 3$; 55: $2^4\cdot 3$; 64: $2^4\cdot 7^2$; 65: $2^4\cdot 7^2$
6: 8: $3^5\cdot 7$; 21: $2^4\cdot 7^2$; 25: $2^2\cdot 3^4\cdot 7^4$; 35: $2^4\cdot 3^3\cdot 7$; 45: $2^2\cdot 3\cdot 7^2$; 50: $2^4\cdot 7$; 54: $3^2\cdot 13$; 56: 7; 58: $2^2\cdot 3^4$; 62: 7^2; 70: $2^4\cdot 13^2$; 74: 2^4
7: 9: $3^5\cdot 7$; 20: $2^4\cdot 7^2$; 24: $2^2\cdot 3^4\cdot 7^4$; 34: $2^4\cdot 3^3\cdot 7$; 44: $2^2\cdot 3\cdot 7^2$; 51: $2^4\cdot 7$; 55: $3^2\cdot 13$; 57: 7; 59: $2^2\cdot 3^4$; 63: 7^2; 71: $2^4\cdot 13^2$; 75: 2^4
8: 10: $3\cdot 7$; 12: $3\cdot 7$; 14: $2^2\cdot 7^2$; 17: $2^2\cdot 7\cdot 13$; 19: $2^2\cdot 5^2\cdot 7\cdot 13$; 29: $2^2\cdot 3\cdot 5^2\cdot 19$; 36: $2^2\cdot 3\cdot 13\cdot 19$; 46: $2^2\cdot 7^2\cdot 19$; 49: $5^2\cdot 13$; 57: $7^2\cdot 19$; 69: $2^4\cdot 3^2$; 72: $2^2\cdot 5^2$; 75: 2^4
9: 11: $3\cdot 7$; 13: $3\cdot 7$; 15: $2^2\cdot 7^2$; 16: $2^2\cdot 7\cdot 13$; 18: $2^2\cdot 5^2\cdot 7\cdot 13$; 28: $2^2\cdot 3\cdot 5^2\cdot 19$; 37: $2^2\cdot 3\cdot 13\cdot 19$; 47: $2^2\cdot 7^2\cdot 19$; 48: $5^2\cdot 13$; 56: $7^2\cdot 19$; 68: $2^4\cdot 3^2$; 73: $2^2\cdot 5^2$; 74: 2^4
10: 25: $2^2\cdot 3^4\cdot 5^2\cdot 7$; 34: $3^4\cdot 7\cdot 19$; 35: $2^2\cdot 3^4\cdot 7$; 42: $2^2\cdot 7\cdot 13$; 44: $2^2\cdot 3\cdot 7^6$; 46: $7^2\cdot 13$; 54: $3\cdot 5^2\cdot 7$; 58: $2^2\cdot 3\cdot 7^2$; 59: $2^2\cdot 3\cdot 13\cdot 19$; 68: $3^4\cdot 13$
11: 24: $2^2\cdot 3^4\cdot 5^2\cdot 7$; 34: $2^2\cdot 3^4\cdot 7$; 35: $3^4\cdot 7\cdot 19$; 43: $2^2\cdot 7\cdot 13$; 45: $2^2\cdot 3\cdot 7^6$; 47: $7^2\cdot 13$; 55: $3\cdot 5^2\cdot 7$; 58: $2^2\cdot 3\cdot 13\cdot 19$; 59: $2^2\cdot 3\cdot 7^2$; 69: $3^4\cdot 13$
12: 24: 3^6; 27: $2^4\cdot 3^6\cdot 13$; 31: $7\cdot 19^2$; 32: $2^2\cdot 7$; 34: 3^3; 57: 13; 62: $3^2\cdot 19$; 63: $3^3\cdot 7$; 68: $2^4\cdot 3^2$; 69: $3^2\cdot 7$
13: 25: 3^6; 26: $2^4\cdot 3^6\cdot 13$; 30: $7\cdot 19^2$; 33: $2^2\cdot 7$; 35: 3^3; 56: 13; 62: $3^3\cdot 7$; 63: $3^2\cdot 19$; 68: $3^2\cdot 7$; 69: $2^4\cdot 3^2$
14: 15: $2^{16}\cdot 3^4$; 18: $2^8\cdot 3^2\cdot 7^4$; 20: $2^8\cdot 7^2\cdot 19$; 24: $2^8\cdot 3^8\cdot 7\cdot 19$; 27: $2^8\cdot 3^{10}\cdot 7$; 28: $2^{16}\cdot 7$; 29: $2^{16}\cdot 7$; 33: $2^8\cdot 7$; 35: $2^{14}\cdot 7$; 36: $2^{16}\cdot 3^4\cdot 7$; 37: $2^{16}\cdot 3^2$; 39: $2^8\cdot 3^2\cdot 7^2$; 43: $2^{16}\cdot 3^4\cdot 7$; 45: $2^{16}\cdot 3^2\cdot 7^4$; 49: $2^8\cdot 3^4\cdot 7$; 50: $2^8\cdot 3^6$; 51: $2^8\cdot 3^4\cdot 7$; 52: $2^8\cdot 7$; 55: $2^{12}\cdot 3^2\cdot 7\cdot 19$; 62: $2^{12}\cdot 3^4\cdot 7$; 65: $2^{16}\cdot 3^4\cdot 7$; 67: $2^{18}\cdot 3^4\cdot 7$; 73: $2^8\cdot 7^2$; 75: $2^8\cdot 3^2$
15: 19: $2^8\cdot 3^2\cdot 7^4$; 21: $2^8\cdot 7^2\cdot 19$; 25: $2^8\cdot 3^8\cdot 7\cdot 19$; 26: $2^8\cdot 3^{10}\cdot 7$; 28: $2^{16}\cdot 7$; 29: $2^{16}\cdot 7$; 32: $2^8\cdot 7$; 34: $2^{14}\cdot 7$; 36: $2^{16}\cdot 3^2$; 37: $2^{16}\cdot 3^4\cdot 7$; 38: $2^8\cdot 3^2\cdot 7^2$; 42: $2^{16}\cdot 3^4\cdot 7$; 44: $2^{16}\cdot 3^2\cdot 7^4$; 48: $2^8\cdot 3^4\cdot 7$; 50: $2^8\cdot 3^4\cdot 7$; 51: $2^8\cdot 3^6$; 53: $2^8\cdot 7$; 54: $2^{12}\cdot 3^2\cdot 7\cdot 19$; 63: $2^{12}\cdot 3^4\cdot 7$; 64: $2^{16}\cdot 3^4\cdot 7$; 66: $2^{18}\cdot 3^4\cdot 7$; 72: $2^8\cdot 7^2$; 74: $2^8\cdot 3^2$
16: 18: $3^2\cdot 7\cdot 13$; 21: $2^2\cdot 7\cdot 13$; 22: $2^6\cdot 3^3$; 24: $3^6\cdot 7$; 35: $2^8\cdot 7$; 37: $2^4\cdot 3^3\cdot 13$; 40: $2^6\cdot 3^4$; 64: $2^4\cdot 3^3\cdot 13$; 70: $2^8\cdot 3^2$
17: 19: $3^2\cdot 7\cdot 13$; 20: $2^2\cdot 7\cdot 13$; 23: $2^6\cdot 3^3$; 25: $3^6\cdot 7$; 34: $2^8\cdot 7$; 36: $2^4\cdot 3^3\cdot 13$; 41: $2^6\cdot 3^4$; 65: $2^4\cdot 3^3\cdot 13$; 71: $2^8\cdot 3^2$
18: 19: $2^2\cdot 3^4\cdot 7^2$; 21: $2^4\cdot 7\cdot 13\cdot 19$; 29: $2^8\cdot 7^2$; 48: $3^2\cdot 7\cdot 13$; 54: $3^2\cdot 7\cdot 19$; 56: 7; 58: $3^2\cdot 7\cdot 13$; 60: $2^4\cdot 3^2\cdot 7^2\cdot 19$; 61: $2^2\cdot 3^2\cdot 7^5$; 64: $2^2\cdot 3^4\cdot 7\cdot 19$; 65: $2^2\cdot 3^4\cdot 7^3$; 69: 2^6; 71: $3^2\cdot 7\cdot 19^2$; 73: $5^2\cdot 7^2$
19: 20: $2^4\cdot 7\cdot 13\cdot 19$; 28: $2^8\cdot 7^2$; 49: $3^2\cdot 7\cdot 13$; 55: $3^2\cdot 7\cdot 19$; 57: 7; 59: $3^2\cdot 7\cdot 13$; 60: $2^2\cdot 3^2\cdot 7^5$; 61: $2^4\cdot 3^2\cdot 7^2\cdot 19$; 64: $2^2\cdot 3^4\cdot 7^3$; 65: $2^2\cdot 3^4\cdot 7\cdot 19$; 68: 2^6; 70: $3^2\cdot 7\cdot 19^2$; 72: $5^2\cdot 7^2$
20: 22: $2^2\cdot 7^2$; 24: $2^2\cdot 3^4\cdot 7\cdot 19$; 26: $2^6\cdot 3^4\cdot 19$; 31: $2^6\cdot 19$; 45: $2^6\cdot 7^2\cdot 13$; 48: $2^4\cdot 19$; 57: $2^2\cdot 7\cdot 13$; 64: $2^2\cdot 7\cdot 13\cdot 19$
21: 23: $2^2\cdot 7^2$; 25: $2^2\cdot 3^4\cdot 7\cdot 19$; 27: $2^6\cdot 3^4\cdot 19$; 30: $2^6\cdot 19$; 44: $2^6\cdot 7^2\cdot 13$; 49: $2^4\cdot 19$; 56: $2^2\cdot 7\cdot 13$; 65: $2^2\cdot 7\cdot 13\cdot 19$
22: 25: $2^6\cdot 3^7\cdot 7^5$; 29: $2^{12}\cdot 5^2\cdot 19$; 34: $2^{14}\cdot 7\cdot 19$; 40: $2^{10}\cdot 3^6\cdot 19$; 41: $2^6\cdot 3^{11}$; 49: $2^6\cdot 3^2\cdot 5^2\cdot 7\cdot 13$; 57: $2^{14}\cdot 19$; 66: $2^{16}\cdot 3^4\cdot 7\cdot 19$; 71: $2^{14}\cdot 3^6$; 73: $2^8\cdot 7^2$; 74: $2^8\cdot 3^4\cdot 13$; 75: $2^6\cdot 3^4\cdot 19$
23: 24: $2^6\cdot 3^7\cdot 7^5$; 28: $2^{12}\cdot 5^2\cdot 19$; 35: $2^{14}\cdot 7\cdot 19$; 40: $2^6\cdot 3^{11}$; 41: $2^{10}\cdot 3^6\cdot 19$; 48: $2^6\cdot 3^2\cdot 5^2\cdot 7\cdot 13$; 56: $2^{14}\cdot 19$; 67: $2^{16}\cdot 3^4\cdot 7\cdot 19$; 70: $2^{14}\cdot 3^6$; 72: $2^8\cdot 7^2$; 74: $2^6\cdot 3^4\cdot 19$; 75: $2^8\cdot 3^4\cdot 13$
24: 25: $2^{10}\cdot 3^{18}$; 34: $3^{10}\cdot 7^4$; 36: $2^2\cdot 3^{13}\cdot 7^2$; 39: $3^{12}\cdot 7^4$; 48: 3^{12}; 49: $3^{12}\cdot 7\cdot 13$; 54: $3^{13}\cdot 7$; 55: $3^{12}\cdot 5^2\cdot 7\cdot 19$; 58: $2^2\cdot 3^{13}\cdot 7^2$; 60: $2^2\cdot 3^{15}\cdot 13$; 61: $2^2\cdot 3^{12}\cdot 7^5$
25: 35: $3^{10}\cdot 7^4$; 37: $2^2\cdot 3^{13}\cdot 7^2$; 38: $3^{12}\cdot 7^4$; 48: $3^{12}\cdot 7\cdot 13$; 49: 3^{12}; 54: $3^{12}\cdot 5^2\cdot 7\cdot 19$; 55: $3^{13}\cdot 7$; 59: $2^2\cdot 3^{13}\cdot 7^2$; 60: $2^2\cdot 3^{12}\cdot 7^5$; 61: $2^2\cdot 3^{15}\cdot 13$
26: 33: $2^8\cdot 3^{10}$; 35: $3^{10}\cdot 7^4$; 37: $2^2\cdot 3^{13}\cdot 7\cdot 19$; 43: $2^2\cdot 3^{14}\cdot 7^2\cdot 13$; 53: $3^{10}\cdot 7^4$; 54: $3^{12}\cdot 13$; 56: $3^{11}\cdot 7\cdot 13^2$; 62: $2^6\cdot 3^{14}\cdot 19$
27: 32: $2^8\cdot 3^{10}$; 34: $3^{10}\cdot 7^4$; 36: $2^2\cdot 3^{13}\cdot 7\cdot 19$; 42: $2^2\cdot 3^{14}\cdot 7^2\cdot 13$; 52: $3^{10}\cdot 7^4$; 55: $3^{12}\cdot 13$; 57: $3^{11}\cdot 7\cdot 13^2$; 63: $2^6\cdot 3^{14}\cdot 19$
28: 35: $2^4\cdot 3^3\cdot 19$; 44: $2^8\cdot 3\cdot 7^2$; 52: 2^6; 54: $2^4\cdot 3^2\cdot 7^2$; 58: $2^2\cdot 3^4$; 59: $2^2\cdot 3\cdot 7^2$; 60: $2^4\cdot 7^2\cdot 19$; 71: $2^4\cdot 19^2$
29: 34: $2^4\cdot 3^3\cdot 19$; 45: $2^8\cdot 3\cdot 7^2$; 53: 2^6; 55: $2^4\cdot 3^2\cdot 7^2$; 58: $2^2\cdot 3\cdot 7^2$; 59: $2^2\cdot 3^4$; 61: $2^4\cdot 7^2\cdot 19$; 70: $2^4\cdot 19^2$
30: 50: $2^6\cdot 19$; 61: $2^6\cdot 7\cdot 19^2$; 65: $2^6\cdot 7\cdot 19^2$
31: 51: $2^6\cdot 19$; 60: $2^6\cdot 7\cdot 19^2$; 64: $2^6\cdot 7\cdot 19^2$
32: 33: $2^4\cdot 3^4$; 34: $7\cdot 19$; 36: $2^2\cdot 7$; 54: $7^2\cdot 13$; 63: $2^2\cdot 5^2\cdot 7\cdot 19$; 69: $2^2\cdot 7\cdot 13$; 70: $5^2\cdot 13^2$; 72: $2^2\cdot 7$; 75: 2^6
33: 35: $7\cdot 19$; 37: $2^2\cdot 7$; 55: $7^2\cdot 13$; 62: $2^2\cdot 5^2\cdot 7\cdot 19$; 68: $2^2\cdot 7\cdot 13$; 71: $5^2\cdot 13^2$; 73: $2^2\cdot 7$; 74: 2^6
34: 45: $2^{18}\cdot 3^3\cdot 7$; 46: $2^{14}\cdot 7^4$; 53: $2^6\cdot 7^4$; 54: $2^{12}\cdot 3^2\cdot 7^2$; 55: $2^{14}\cdot 3^4$; 57: $2^{14}\cdot 7$; 59: $2^6\cdot 3^3\cdot 7^3\cdot 19$; 60: $2^{14}\cdot 7^4$; 63: $2^{14}\cdot 19$; 66: $2^{16}\cdot 7^3$; 71: 2^{22}
35: 44: $2^{18}\cdot 3^3\cdot 7$; 47: $2^{14}\cdot 7^4$; 52: $2^6\cdot 7^4$; 54: $2^{14}\cdot 3^4$; 55: $2^{12}\cdot 3^2\cdot 7^2$; 56: $2^{14}\cdot 7$; 58: $2^6\cdot 3^3\cdot 7^3\cdot 19$; 61: $2^{14}\cdot 7^4$; 62: $2^{14}\cdot 19$; 67: $2^{16}\cdot 7^3$; 70: 2^{22}
36: 38: $2^8\cdot 3^2\cdot 5^2\cdot 7$; 39: $2^8\cdot 3^2\cdot 7^2\cdot 19^2$; 40: $2^8\cdot 3^5\cdot 7^2\cdot 19$; 42: $2^{16}\cdot 3^2\cdot 5^2\cdot 7$; 45: $2^{16}\cdot 3^5\cdot 7^2$; 46: $2^{12}\cdot 3^2\cdot 7^4$; 49: $2^8\cdot 3^2\cdot 7^2\cdot 13$; 57: $2^{12}\cdot 19^2$; 59: $2^8\cdot 3^4\cdot 7\cdot 13\cdot 19$; 60: $2^{14}\cdot 3^4\cdot 7\cdot 19$; 61: $2^{14}\cdot 3^2\cdot 7^2\cdot 19$; 70: $2^{14}\cdot 3^6$; 75: $2^8\cdot 3^2\cdot 19^2$
37: 38: $2^8\cdot 3^2\cdot 7^2\cdot 19^2$; 39: $2^8\cdot 3^2\cdot 5^2\cdot 7$; 41: $2^8\cdot 3^5\cdot 7^2\cdot 19$; 43: $2^{16}\cdot 3^2\cdot 5^2\cdot 7$; 44: $2^{16}\cdot 3^5\cdot 7^2$; 47: $2^{12}\cdot 3^2\cdot 7^4$; 48: $2^8\cdot 3^2\cdot 7^2\cdot 13$; 56: $2^{12}\cdot 19^2$; 58: $2^8\cdot 3^4\cdot 7\cdot 13\cdot 19$; 60: $2^{14}\cdot 3^2\cdot 7^2\cdot 19$; 61: $2^{14}\cdot 3^4\cdot 7\cdot 19$; 71: $2^{14}\cdot 3^6$; 74: $2^8\cdot 3^2\cdot 19^2$
38: 39: $2^4\cdot 3^4$; 56: $7^2\cdot 19$; 62: $3^2\cdot 19^2$; 73: $5^2\cdot 7\cdot 19$
39: 57: $7^2\cdot 19$; 63: $3^2\cdot 19^2$; 72: $5^2\cdot 7\cdot 19$
40: 50: $2^4\cdot 3^3\cdot 7$; 63: $3^3\cdot 7$; 65: $2^2\cdot 3^7\cdot 7^2$; 70: $2^6\cdot 3^7$
41: 51: $2^4\cdot 3^3\cdot 7$; 62: $3^3\cdot 7$; 64: $2^2\cdot 3^7\cdot 7^2$; 71: $2^6\cdot 3^7$
42: 61: $2^4\cdot 3^6\cdot 7$; 71: $2^4\cdot 3^4\cdot 5^2$
43: 60: $2^4\cdot 3^6\cdot 7$; 70: $2^4\cdot 3^4\cdot 5^2$
44: 45: $2^{18}\cdot 3^6\cdot 7^2$; 50: $2^{10}\cdot 3^3\cdot 7\cdot 19^2$; 56: $2^{14}\cdot 7\cdot 13$; 57: $2^{12}\cdot 7^4$; 61: $2^{14}\cdot 3^2\cdot 7^3\cdot 19$; 65: $2^{16}\cdot 3^3\cdot 7\cdot 13^2$
45: 51: $2^{10}\cdot 3^3\cdot 7\cdot 19^2$; 56: $2^{12}\cdot 7^4$; 57: $2^{14}\cdot 7\cdot 13$; 60: $2^{14}\cdot 3^2\cdot 7^3\cdot 19$; 64: $2^{16}\cdot 3^3\cdot 7\cdot 13^2$
46: 49: $3^2\cdot 13^2$; 59: $3^2\cdot 7\cdot 13$; 60: $2^6\cdot 3^2\cdot 7\cdot 13$; 70: $2^6\cdot 3^2\cdot 19^2$
47: 48: $3^2\cdot 13^2$; 58: $3^2\cdot 7\cdot 13$; 61: $2^6\cdot 3^2\cdot 7\cdot 13$; 71: $2^6\cdot 3^2\cdot 19^2$
48: 54: $3^2\cdot 13$; 55: $3^2\cdot 7\cdot 19$; 64: $2^2\cdot 3^2$; 70: 3^4
49: 54: $3^2\cdot 7\cdot 19$; 55: $3^2\cdot 13$; 65: $2^2\cdot 3^2$; 71: 3^4
50: 54: $3^3\cdot 7$; 60: $3^8\cdot 7$; 63: $3^2\cdot 19^2$; 68: $2^2\cdot 7\cdot 19$; 70: $2^2\cdot 3^6$; 71: 3^4; 72: $2^4\cdot 19$
51: 55: $3^3\cdot 7$; 61: $3^8\cdot 7$; 62: $3^2\cdot 19^2$; 69: $2^2\cdot 7\cdot 19$; 70: 3^4; 71: $2^2\cdot 3^6$; 73: $2^4\cdot 19$
52: 56: 7; 60: 7; 64: 2^2; 69: 2^2
53: 57: 7; 61: 7; 65: 2^2; 68: 2^2
54: 57: $2^{12}\cdot 7$; 66: $2^{14}\cdot 3^2\cdot 7\cdot 19$; 69: $2^6\cdot 3^2\cdot 13$; 71: $2^{12}\cdot 3^2$
55: 56: $2^{12}\cdot 7$; 67: $2^{14}\cdot 3^2\cdot 7\cdot 19$;

$68\colon 2^6 \cdot 3^2 \cdot 13$; $70\colon 2^{12} \cdot 3^2$ **56**: $61\colon 2^6 \cdot 19$; $75\colon 2^4$ **57**: $60\colon 2^6 \cdot 19$; $74\colon 2^4$ **58**: $74\colon 3^2 \cdot 19$ **59**: $75\colon 3^2 \cdot 19$
60: $62\colon 2^{14} \cdot 3^4 \cdot 19$; $63\colon 2^{16} \cdot 3^2 \cdot 7$; $66\colon 2^{16} \cdot 3^2 \cdot 7^4$; $67\colon 2^{16} \cdot 3^4 \cdot 19$; $71\colon 2^{14} \cdot 3^4 \cdot 19$; $72\colon 2^6 \cdot 7^4$; $74\colon 2^6 \cdot 3^2 \cdot 19$
61: $62\colon 2^{16} \cdot 3^2 \cdot 7$; $63\colon 2^{14} \cdot 3^4 \cdot 19$; $66\colon 2^{16} \cdot 3^4 \cdot 19$; $67\colon 2^{16} \cdot 3^2 \cdot 7^4$; $70\colon 2^{14} \cdot 3^4 \cdot 19$; $73\colon 2^6 \cdot 7^4$; $75\colon 2^6 \cdot 3^2 \cdot 19$
62: $64\colon 2^4 \cdot 3^3 \cdot 19$; $70\colon 2^6 \cdot 3^6$; $71\colon 2^6 \cdot 3^2 \cdot 5^2$ **63**: $65\colon 2^4 \cdot 3^3 \cdot 19$; $70\colon 2^6 \cdot 3^2 \cdot 5^2$; $71\colon 2^6 \cdot 3^6$ **64**: $65\colon 2^{16} \cdot 3^6$;
$66\colon 2^{18} \cdot 3^3 \cdot 7^2$; $67\colon 2^{16} \cdot 3^3 \cdot 7$; $71\colon 2^{14} \cdot 3^4 \cdot 19$; $74\colon 2^8 \cdot 3^4$ **65**: $66\colon 2^{16} \cdot 3^3 \cdot 7$; $67\colon 2^{18} \cdot 3^3 \cdot 7^2$; $70\colon 2^{14} \cdot 3^4 \cdot 19$;
$75\colon 2^8 \cdot 3^4$ **66**: $70\colon 2^{14} \cdot 3^2 \cdot 5^2 \cdot 7$; $71\colon 2^{14} \cdot 3^5$ **67**: $70\colon 2^{14} \cdot 3^5$; $71\colon 2^{14} \cdot 3^2 \cdot 5^2 \cdot 7$ **68**: $69\colon 2^2 \cdot 3^6$; $72\colon 2^2$
69: $73\colon 2^2$ **70**: $71\colon 2^{16} \cdot 3^4$; $72\colon 2^6 \cdot 5^2 \cdot 7$; $74\colon 2^8 \cdot 3^2$; $75\colon 2^6 \cdot 3^2 \cdot 5^4$ **71**: $73\colon 2^6 \cdot 5^2 \cdot 7$; $74\colon 2^6 \cdot 3^2 \cdot 5^4$; $75\colon 2^8 \cdot 3^2$
72: $75\colon 2^8$ **73**: $74\colon 2^8$ **74**: $75\colon 3^4$

Table 3: the terminal residue classes $x \equiv a$, $y \equiv b \pmod{p^k}$ of the local computation for the cubic graph, grouped by p, k and the value of B_p.

p	k	B_p	classes (a, b)
2	2	1	$(2,3)$
2	3	1	$(0,5)$, $(1,3)$, $(1,7)$, $(4,1)$, $(5,1)$, $(5,5)$
3	3	2	$(0,14)$, $(6,11)$, $(15,20)$, $(24,2)$
3	4	2	$(3,35)$, $(9,50)$, $(12,71)$, $(18,59)$, $(21,26)$, $(30,62)$, $(36,77)$, $(39,17)$, $(45,5)$, $(48,53)$, $(57,8)$, $(63,23)$, $(66,44)$, $(72,32)$, $(75,80)$
5	1	0	$(0,3)$, $(2,1)$, $(2,3)$, $(3,4)$, $(4,2)$
7	2	1	$(0,25)$, $(2,16)$, $(7,5)$, $(7,40)$, $(7,47)$, $(9,23)$, $(23,37)$, $(25,41)$, $(30,44)$, $(32,20)$, $(35,39)$
7	2	2	$(8,31)$, $(15,3)$, $(19,15)$, $(22,24)$, $(29,45)$, $(31,9)$, $(45,30)$, $(47,29)$
7	3	1	$(4,251)$, $(7,67)$, $(11,279)$, $(14,11)$, $(21,4)$, $(28,46)$, $(37,296)$, $(39,97)$, $(42,277)$, $(44,303)$, $(46,321)$, $(53,104)$, $(60,132)$, $(63,305)$, $(65,30)$, $(67,62)$, $(70,298)$, $(77,340)$, $(86,2)$, $(88,293)$, $(91,228)$, $(93,9)$, $(95,174)$, $(102,300)$, $(105,26)$, $(105,75)$, $(105,124)$, $(105,173)$, $(105,222)$, $(105,271)$, $(105,312)$, $(105,320)$, $(109,328)$, $(112,256)$, $(114,79)$, $(116,258)$, $(119,249)$, $(126,291)$, $(135,51)$, $(137,146)$, $(140,179)$, $(142,58)$, $(144,27)$, $(151,153)$, $(154,33)$, $(154,82)$, $(154,131)$, $(154,180)$, $(154,229)$, $(154,263)$, $(154,278)$, $(154,327)$, $(158,181)$, $(161,207)$, $(163,128)$, $(165,111)$, $(168,200)$, $(175,242)$, $(184,100)$, $(186,342)$, $(189,130)$, $(191,107)$, $(193,223)$, $(200,6)$, $(203,214)$, $(207,34)$, $(210,158)$, $(212,177)$, $(214,307)$, $(217,151)$, $(224,193)$, $(233,149)$, $(235,195)$, $(238,81)$, $(240,156)$, $(242,76)$, $(249,202)$, $(252,12)$, $(252,19)$, $(252,61)$, $(252,68)$, $(252,110)$, $(252,117)$, $(252,159)$, $(252,165)$, $(252,166)$, $(252,208)$, $(252,215)$, $(252,257)$, $(252,264)$, $(252,306)$, $(252,313)$, $(256,230)$, $(259,109)$, $(261,226)$, $(263,160)$, $(266,102)$, $(273,144)$, $(282,198)$, $(284,48)$, $(287,32)$, $(289,205)$, $(291,272)$, $(298,55)$, $(301,116)$, $(305,83)$, $(308,60)$, $(310,275)$, $(312,13)$, $(315,53)$, $(322,95)$, $(333,244)$, $(336,326)$, $(338,254)$, $(340,125)$
7	3	2	$(1,304)$, $(5,155)$, $(10,2)$, $(12,36)$, $(17,135)$, $(24,219)$, $(26,190)$, $(33,120)$, $(36,66)$, $(38,240)$, $(40,295)$, $(43,283)$, $(50,108)$, $(54,8)$, $(59,247)$, $(61,232)$, $(66,37)$, $(73,121)$, $(75,43)$, $(82,316)$, $(85,213)$, $(87,142)$, $(89,148)$, $(92,87)$, $(99,255)$, $(108,149)$, $(110,85)$, $(115,282)$, $(122,23)$, $(124,239)$, $(131,169)$, $(134,17)$, $(136,44)$, $(138,1)$, $(141,234)$, $(148,59)$, $(157,51)$, $(159,281)$, $(164,184)$, $(171,268)$, $(173,92)$, $(180,22)$, $(183,164)$, $(185,289)$, $(187,197)$, $(190,38)$, $(197,206)$, $(206,296)$, $(208,134)$, $(213,86)$, $(220,170)$, $(222,288)$, $(229,218)$, $(232,311)$, $(234,191)$, $(236,50)$, $(239,185)$, $(246,10)$, $(255,198)$, $(257,330)$, $(262,331)$, $(269,72)$, $(271,141)$, $(278,71)$, $(281,115)$, $(283,93)$, $(285,246)$, $(295,157)$, $(304,100)$, $(306,183)$, $(311,233)$, $(318,317)$, $(320,337)$, $(327,267)$, $(330,262)$, $(332,338)$, $(334,99)$, $(337,136)$
7	4	1	$(16,1010)$, $(18,895)$, $(56,1733)$, $(331,1619)$, $(359,1353)$, $(361,2267)$, $(399,1390)$, $(674,1962)$, $(702,1696)$, $(704,1238)$, $(1017,2305)$, $(1045,2039)$, $(1047,209)$, $(1085,704)$, $(1360,247)$, $(1388,2382)$, $(1390,1581)$, $(1428,361)$, $(1703,590)$, $(1731,324)$, $(1733,552)$, $(1771,18)$, $(2046,933)$, $(2074,667)$, $(2076,1924)$, $(2114,2076)$, $(2389,1276)$

p	k	B_p	classes (a, b)
7	4	2	$(3, 849)$, $(52, 1094)$, $(101, 1339)$, $(103, 1576)$, $(150, 1584)$, $(152, 1086)$, $(199, 1829)$, $(201, 596)$, $(250, 106)$, $(288, 2047)$, $(297, 2319)$, $(299, 2017)$, $(346, 163)$, $(395, 408)$, $(444, 653)$, $(446, 547)$, $(493, 898)$, $(495, 57)$, $(542, 1143)$, $(544, 1968)$, $(593, 1478)$, $(631, 675)$, $(640, 1633)$, $(642, 988)$, $(689, 1878)$, $(738, 2123)$, $(787, 2368)$, $(789, 1919)$, $(836, 212)$, $(838, 1429)$, $(885, 457)$, $(887, 939)$, $(936, 449)$, $(974, 1704)$, $(983, 947)$, $(985, 2360)$, $(1032, 1192)$, $(1081, 1437)$, $(1130, 1682)$, $(1132, 890)$, $(1179, 1927)$, $(1181, 400)$, $(1228, 2172)$, $(1230, 2311)$, $(1279, 1821)$, $(1317, 332)$, $(1326, 261)$, $(1328, 1331)$, $(1375, 506)$, $(1424, 751)$, $(1473, 996)$, $(1475, 2262)$, $(1522, 1241)$, $(1524, 1772)$, $(1571, 1486)$, $(1573, 1282)$, $(1622, 792)$, $(1660, 1361)$, $(1669, 1976)$, $(1671, 302)$, $(1718, 2221)$, $(1767, 65)$, $(1816, 310)$, $(1818, 1233)$, $(1865, 555)$, $(1867, 743)$, $(1914, 800)$, $(1916, 253)$, $(1965, 2164)$, $(2003, 2390)$, $(2012, 1290)$, $(2014, 1674)$, $(2061, 1535)$, $(2110, 1780)$, $(2159, 2025)$, $(2161, 204)$, $(2208, 2270)$, $(2210, 2115)$, $(2257, 114)$, $(2259, 1625)$, $(2308, 1135)$, $(2346, 1018)$, $(2355, 604)$, $(2357, 645)$
7	5	1	$(742, 15453)$, $(3143, 13052)$, $(5544, 10651)$, $(7945, 8250)$, $(10346, 5849)$, $(12747, 3448)$, $(15148, 1047)$
7	5	2	$(248, 16480)$, $(591, 1388)$, $(934, 3103)$, $(1277, 4818)$, $(1620, 6533)$, $(2306, 9963)$, $(2649, 11678)$, $(2992, 13393)$, $(3335, 15108)$, $(3678, 16)$, $(4021, 1731)$, $(4707, 5161)$, $(5050, 6876)$, $(5393, 8591)$, $(5736, 10306)$, $(6079, 12021)$, $(6422, 13736)$, $(7108, 359)$, $(7451, 2074)$, $(7794, 3789)$, $(8137, 5504)$, $(8480, 7219)$, $(8823, 8934)$, $(9509, 12364)$, $(9852, 14079)$, $(10195, 15794)$, $(10538, 702)$, $(10881, 2417)$, $(11224, 4132)$, $(11910, 7562)$, $(12253, 9277)$, $(12596, 10992)$, $(12939, 12707)$, $(13282, 14422)$, $(13625, 16137)$, $(13968, 1045)$, $(14311, 2760)$, $(14654, 4475)$, $(14997, 6190)$, $(15340, 7905)$, $(15683, 9620)$, $(16026, 11335)$, $(16369, 13050)$, $(16712, 14765)$
7	6	2	$(1963, 58669)$, $(4364, 70674)$, $(6765, 82679)$, $(9166, 94684)$, $(11567, 106689)$, $(18770, 25055)$, $(21171, 37060)$, $(23572, 49065)$, $(25973, 61070)$, $(28374, 73075)$, $(35577, 109090)$, $(37978, 3446)$, $(40379, 15451)$, $(42780, 27456)$, $(45181, 39461)$, $(52384, 75476)$, $(54785, 87481)$, $(57186, 99486)$, $(59587, 111491)$, $(61988, 5847)$, $(69191, 41862)$, $(71592, 53867)$, $(73993, 65872)$, $(76394, 77877)$, $(78795, 89882)$, $(85998, 8248)$, $(88399, 20253)$, $(90800, 32258)$, $(93201, 44263)$, $(95602, 56268)$, $(102805, 92283)$, $(105206, 104288)$, $(107607, 116293)$, $(110008, 10649)$, $(112409, 22654)$
13	1	0	$(0, 7)$, $(10, 8)$
13	2	0	$(3, 84)$, $(4, 9)$, $(4, 32)$, $(8, 50)$, $(9, 30)$, $(11, 93)$, $(16, 32)$, $(17, 9)$, $(17, 58)$, $(21, 141)$, $(24, 80)$, $(29, 149)$, $(30, 9)$, $(30, 84)$, $(34, 63)$, $(35, 30)$, $(37, 67)$, $(42, 97)$, $(43, 9)$, $(43, 50)$, $(43, 110)$, $(47, 154)$, $(48, 30)$, $(50, 54)$, $(55, 45)$, $(56, 9)$, $(56, 136)$, $(60, 76)$, $(61, 30)$, $(63, 41)$, $(68, 162)$, $(69, 9)$, $(69, 162)$, $(73, 167)$, $(74, 30)$, $(76, 28)$, $(81, 110)$, $(82, 9)$, $(82, 19)$, $(86, 89)$, $(87, 30)$, $(89, 15)$, $(94, 58)$, $(95, 9)$, $(95, 45)$, $(99, 11)$, $(100, 30)$, $(107, 6)$, $(108, 9)$, $(108, 71)$, $(108, 89)$, $(112, 102)$, $(113, 30)$, $(115, 158)$, $(121, 9)$, $(121, 97)$, $(125, 24)$, $(126, 30)$, $(128, 145)$, $(133, 71)$, $(134, 9)$, $(134, 37)$, $(134, 123)$, $(138, 115)$, $(139, 30)$, $(141, 132)$, $(146, 19)$, $(147, 9)$, $(147, 11)$, $(147, 149)$, $(151, 37)$, $(152, 30)$, $(154, 119)$, $(159, 136)$, $(160, 6)$, $(160, 9)$, $(164, 128)$, $(165, 30)$, $(167, 106)$

p	k	B_p	classes (a, b)
13	3	0	$(4, 1480)$, $(17, 271)$, $(22, 706)$, $(30, 921)$, $(56, 1207)$, $(69, 843)$, $(82, 141)$, $(95, 1298)$, $(102, 1523)$, $(120, 2151)$, $(121, 401)$, $(160, 2013)$, $(173, 1142)$, $(186, 2130)$, $(191, 706)$, $(199, 583)$, $(225, 869)$, $(238, 505)$, $(251, 2000)$, $(264, 960)$, $(271, 1354)$, $(289, 1475)$, $(290, 63)$, $(329, 1675)$, $(342, 804)$, $(355, 1792)$, $(360, 706)$, $(368, 245)$, $(394, 531)$, $(407, 167)$, $(420, 1662)$, $(433, 622)$, $(440, 1185)$, $(458, 799)$, $(459, 1922)$, $(498, 1337)$, $(511, 466)$, $(524, 1454)$, $(529, 706)$, $(537, 2104)$, $(563, 193)$, $(576, 2026)$, $(589, 1324)$, $(602, 284)$, $(609, 1016)$, $(627, 123)$, $(628, 1584)$, $(667, 999)$, $(680, 128)$, $(693, 1116)$, $(698, 706)$, $(706, 1766)$, $(732, 2052)$, $(745, 1688)$, $(758, 986)$, $(771, 2143)$, $(778, 847)$, $(796, 1644)$, $(797, 1246)$, $(836, 661)$, $(849, 1987)$, $(862, 778)$, $(867, 706)$, $(875, 1428)$, $(901, 1714)$, $(914, 1350)$, $(927, 648)$, $(940, 1805)$, $(947, 678)$, $(965, 968)$, $(966, 908)$, $(1005, 323)$, $(1018, 1649)$, $(1031, 440)$, $(1036, 706)$, $(1044, 1090)$, $(1070, 1376)$, $(1083, 1012)$, $(1096, 310)$, $(1109, 1467)$, $(1116, 509)$, $(1134, 292)$, $(1135, 570)$, $(1174, 2182)$, $(1187, 1311)$, $(1200, 102)$, $(1205, 706)$, $(1213, 752)$, $(1239, 1038)$, $(1252, 674)$, $(1265, 2169)$, $(1278, 1129)$, $(1285, 340)$, $(1303, 1813)$, $(1304, 232)$, $(1343, 1844)$, $(1356, 973)$, $(1369, 1961)$, $(1374, 706)$, $(1382, 414)$, $(1408, 700)$, $(1421, 336)$, $(1434, 1831)$, $(1447, 791)$, $(1454, 171)$, $(1472, 1137)$, $(1473, 2091)$, $(1512, 1506)$, $(1525, 635)$, $(1538, 1623)$, $(1543, 706)$, $(1551, 76)$, $(1577, 362)$, $(1590, 2195)$, $(1603, 1493)$, $(1616, 453)$, $(1623, 2)$, $(1641, 461)$, $(1642, 1753)$, $(1681, 1168)$, $(1694, 297)$, $(1707, 1285)$, $(1712, 706)$, $(1720, 1935)$, $(1746, 24)$, $(1759, 1857)$, $(1772, 1155)$, $(1785, 115)$, $(1792, 2030)$, $(1810, 1982)$, $(1811, 1415)$, $(1850, 830)$, $(1863, 2156)$, $(1876, 947)$, $(1881, 706)$, $(1889, 1597)$, $(1915, 1883)$, $(1928, 1519)$, $(1941, 817)$, $(1954, 1974)$, $(1961, 1861)$, $(1979, 1306)$, $(1980, 1077)$, $(2019, 492)$, $(2032, 1818)$, $(2045, 609)$, $(2050, 706)$, $(2058, 1259)$, $(2084, 1545)$, $(2097, 1181)$, $(2110, 479)$, $(2123, 1636)$, $(2130, 1692)$, $(2148, 630)$, $(2149, 739)$, $(2188, 154)$
19	1	2	$(0, 10)$, $(2, 9)$, $(4, 16)$, $(6, 11)$, $(8, 2)$, $(8, 15)$, $(9, 13)$, $(12, 13)$
19	2	2	$(1, 177)$, $(1, 243)$, $(1, 302)$, $(2, 53)$, $(2, 185)$, $(3, 22)$, $(3, 126)$, $(3, 213)$, $(4, 156)$, $(4, 227)$, $(7, 290)$, $(11, 254)$, $(14, 111)$, $(15, 24)$, $(15, 25)$, $(15, 312)$, $(20, 131)$, $(20, 281)$, $(20, 310)$, $(21, 204)$, $(21, 262)$, $(22, 12)$, $(22, 61)$, $(22, 288)$, $(23, 118)$, $(23, 132)$, $(26, 195)$, $(30, 216)$, $(33, 225)$, $(34, 62)$, $(34, 103)$, $(34, 196)$, $(39, 82)$, $(39, 319)$, $(39, 321)$, $(40, 110)$, $(40, 223)$, $(41, 193)$, $(41, 259)$, $(41, 270)$, $(42, 37)$, $(42, 80)$, $(45, 100)$, $(49, 178)$, $(52, 339)$, $(53, 6)$, $(53, 100)$, $(53, 255)$, $(58, 150)$, $(58, 215)$, $(58, 357)$, $(59, 242)$, $(59, 319)$, $(60, 98)$, $(60, 118)$, $(60, 145)$, $(61, 42)$, $(61, 303)$, $(64, 5)$, $(68, 140)$, $(71, 92)$, $(72, 46)$, $(72, 138)$, $(72, 177)$, $(77, 34)$, $(77, 340)$, $(77, 348)$, $(78, 167)$, $(78, 261)$, $(79, 3)$, $(79, 31)$, $(79, 327)$, $(80, 4)$, $(80, 208)$, $(83, 271)$, $(87, 102)$, $(90, 206)$, $(91, 176)$, $(91, 198)$, $(91, 348)$, $(96, 72)$, $(96, 120)$, $(96, 169)$, $(97, 15)$, $(97, 280)$, $(98, 175)$, $(98, 269)$, $(98, 278)$, $(99, 113)$, $(99, 327)$, $(102, 176)$, $(106, 64)$, $(109, 320)$, $(110, 158)$, $(110, 214)$, $(110, 350)$, $(115, 110)$, $(115, 253)$, $(115, 359)$, $(116, 224)$, $(116, 299)$, $(117, 23)$, $(117, 164)$, $(117, 174)$, $(118, 18)$, $(118, 289)$, $(121, 81)$, $(125, 26)$, $(128, 73)$, $(129, 141)$, $(129, 252)$, $(129, 329)$, $(134, 25)$, $(134, 148)$, $(134, 188)$, $(135, 72)$, $(135, 318)$, $(136, 50)$, $(136, 79)$, $(136, 232)$, $(137, 251)$, $(137, 284)$, $(140, 347)$, $(144, 349)$, $(147, 187)$, $(148, 139)$, $(148, 290)$, $(148, 293)$, $(153, 17)$, $(153, 158)$, $(153, 186)$, $(154, 281)$, $(154, 337)$, $(155, 80)$, $(155, 297)$, $(155, 345)$, $(156, 189)$, $(159, 252)$, $(163, 311)$, $(166, 301)$, $(167, 84)$, $(167, 310)$, $(167, 328)$, $(172, 207)$, $(172, 224)$, $(172, 291)$, $(173, 129)$, $(173, 356)$, $(174, 183)$, $(174, 250)$, $(174, 289)$, $(175, 94)$, $(175, 175)$, $(178, 157)$, $(182, 273)$, $(185, 54)$, $(186, 5)$, $(186, 120)$, $(186, 236)$, $(191, 36)$, $(191, 63)$, $(191, 262)$, $(192, 14)$, $(192, 338)$, $(193, 69)$, $(193, 137)$, $(193, 155)$, $(194, 137)$, $(194, 360)$, $(197, 62)$, $(201, 235)$, $(204, 168)$, $(205, 27)$, $(205, 43)$, $(205, 291)$, $(210, 196)$, $(210, 226)$, $(210, 300)$, $(211, 33)$, $(211, 186)$, $(212, 60)$, $(212, 316)$, $(212, 346)$, $(213, 99)$, $(213, 265)$, $(216, 328)$, $(220, 197)$, $(223, 282)$, $(224, 81)$, $(224, 101)$, $(224, 179)$, $(229, 55)$, $(229, 329)$, $(229, 338)$, $(230, 34)$, $(230, 52)$, $(231, 194)$, $(231, 202)$, $(231, 326)$, $(232, 61)$, $(232, 170)$, $(235, 233)$, $(239, 159)$, $(242, 35)$, $(243, 119)$, $(243, 272)$, $(243, 331)$, $(248, 15)$, $(248, 101)$, $(248, 245)$, $(249, 71)$, $(249, 243)$, $(250, 42)$, $(250, 88)$, $(250, 231)$, $(251, 23)$, $(251, 75)$, $(254, 138)$, $(258, 121)$, $(261, 149)$, $(262, 82)$, $(262, 122)$, $(262, 157)$, $(267, 53)$, $(267, 74)$, $(267, 234)$, $(268, 90)$, $(268, 91)$, $(269, 136)$, $(269, 251)$, $(269, 335)$, $(270, 341)$, $(270, 346)$, $(273, 43)$, $(277, 83)$, $(280, 263)$, $(281, 195)$, $(281, 253)$, $(281, 274)$, $(286, 91)$, $(286, 264)$, $(287, 109)$, $(287, 300)$, $(288, 41)$, $(288, 99)$, $(288, 221)$, $(289, 246)$, $(289, 308)$, $(292, 309)$, $(296, 45)$, $(299, 16)$, $(300, 63)$, $(300, 65)$, $(300, 233)$, $(305, 93)$, $(305, 129)$, $(305, 139)$, $(306, 128)$, $(306, 148)$, $(307, 107)$, $(307, 307)$, $(307, 308)$, $(308, 151)$, $(308, 270)$, $(311, 214)$, $(315, 7)$, $(318, 130)$, $(319, 217)$, $(319, 234)$, $(319, 271)$, $(324, 167)$, $(324, 272)$, $(324, 283)$, $(325, 147)$, $(325, 357)$, $(326, 156)$, $(326, 212)$, $(326, 354)$, $(327, 56)$, $(327, 232)$, $(330, 119)$, $(334, 330)$, $(337, 244)$, $(338, 8)$, $(338, 44)$, $(338, 309)$, $(343, 44)$, $(343, 112)$, $(343, 205)$, $(344, 166)$, $(344, 205)$, $(345, 4)$, $(345, 117)$, $(345, 240)$, $(346, 194)$, $(346, 322)$, $(349, 24)$, $(353, 292)$, $(356, 358)$, $(357, 160)$, $(357, 215)$, $(357, 347)$

p	k	B_p	classes (a,b)
19	3	2	$(156,6711)$, $(286,367)$, $(517,5989)$, $(647,2894)$, $(878,5267)$, $(1008,5421)$, $(1239,4545)$, $(1369,1089)$, $(1600,3823)$, $(1730,3616)$, $(1961,3101)$, $(2091,6143)$, $(2322,2379)$, $(2452,1811)$, $(2683,1657)$, $(2813,4338)$, $(3044,935)$, $(3174,6)$, $(3405,213)$, $(3535,2533)$, $(3766,6350)$, $(3896,5060)$, $(4127,5628)$, $(4257,728)$, $(4488,4906)$, $(4618,3255)$, $(4849,4184)$, $(4979,5782)$, $(5210,3462)$, $(5340,1450)$, $(5571,2740)$, $(5701,3977)$, $(5932,2018)$, $(6062,6504)$, $(6293,1296)$, $(6423,2172)$, $(6654,574)$, $(6784,4699)$
37	1	0	$(0,4)$, $(0,14)$, $(0,19)$, $(1,17)$, $(2,28)$, $(3,1)$, $(3,8)$, $(3,28)$, $(5,14)$, $(7,29)$, $(8,5)$, $(8,12)$, $(8,20)$, $(9,18)$, $(11,9)$, $(11,13)$, $(11,15)$, $(12,1)$, $(12,6)$, $(12,30)$, $(13,14)$, $(13,26)$, $(13,34)$, $(14,16)$, $(15,18)$, $(15,21)$, $(15,35)$, $(17,25)$, $(19,5)$, $(19,14)$, $(19,18)$, $(20,5)$, $(22,12)$, $(22,26)$, $(22,36)$, $(24,3)$, $(26,6)$, $(27,5)$, $(28,16)$, $(31,18)$, $(32,21)$, $(34,17)$, $(35,15)$

6.2 The graph of order 8

Elements of $K = \mathbb{Q}(\zeta)$, $\zeta^4 = -1$, are written as polynomials in ζ of degree at most 3. The generators g_k and their norms:

k	g_k	$N(g_k)$	k	g_k	$N(g_k)$
1	ζ	1	14	$2-\zeta^3$	17
2	$1+\zeta+\zeta^2$	1	15	$-2\zeta^2+\zeta^3$	17
3	$1+\zeta$	2	16	$-2\zeta^2-\zeta^3$	17
4	$-1-\zeta^2-\zeta^3$	3^2	17	$1-2\zeta$	17
5	$-1+\zeta+\zeta^2$	3^2	18	$-1-\zeta-2\zeta^2+\zeta^3$	41
6	$-2+\zeta^2$	5^2	19	$2+\zeta+\zeta^2+\zeta^3$	41
7	$2+\zeta^2$	5^2	20	$-2+\zeta-\zeta^2+\zeta^3$	41
8	$2+\zeta+2\zeta^2$	7^2	21	$1-\zeta+2\zeta^2+\zeta^3$	41
9	$-2+\zeta-2\zeta^2$	7^2	22	$-2-2\zeta+\zeta^2$	73
10	$-\zeta-3\zeta^2+\zeta^3$	11^2	23	$-1+2\zeta+2\zeta^2$	73
11	$3+\zeta+\zeta^3$	11^2	24	$1+2\zeta-2\zeta^2$	73
12	$3+2\zeta^2$	13^2	25	$-2+2\zeta+\zeta^2$	73
13	$3-2\zeta^2$	13^2			

The 77 lines H_i and their constants c_i, as products of the g_k:

i	H_i	c_i
0	$(-1-\zeta^2)x+2y-\zeta-\zeta^3$	$g_2^6g_3^3g_9g_{14}^6$
1	$(-1+\zeta^2)x+2y-\zeta-\zeta^3$	1
2	$(-1+\zeta+\zeta^2+\zeta^3)x+2y-1-\zeta+\zeta^2+\zeta^3$	$g_1^2g_2^7g_3^5g_4^3g_5^5g_{14}^3g_{16}^5$
3	$(-1-\zeta-\zeta^2-\zeta^3)x+2y-1-\zeta-\zeta^2+\zeta^3$	1
4	$(-\zeta+\zeta^2+\zeta^3)x+2y-1$	$g_1^6g_2g_{16}^5$
5	$(\zeta+\zeta^2-\zeta^3)x+2y-1$	1
6	$-\zeta^2x+2y-\zeta$	$g_1^3g_2^7g_3^5g_5^4g_{14}^7$
7	$\zeta^2x+2y+\zeta^3$	1
8	$(\zeta-\zeta^3)x+2y-1-\zeta^2$	$g_1^6g_2^7g_3^2g_4^5g_5g_9^2g_{14}^5$
9	$(\zeta-\zeta^3)x+2y-1+\zeta^2$	1
10	$x+2y+\zeta$	$g_1^4g_2g_3g_9g_{15}^3$
11	$x+2y+\zeta^3$	1
12	$\zeta^2x+y-\zeta$	$g_1^7g_2^5g_3^2g_4g_{14}^3g_{17}^5$
13	$-\zeta^2x+y+\zeta^3$	1
14	$(1+\zeta-\zeta^2+\zeta^3)x+2y-1+\zeta+\zeta^2-\zeta^3$	$g_1^5g_2^5g_3^5g_5^2g_{14}^7$
15	$(1+\zeta+\zeta^2+\zeta^3)x+2y-1-\zeta-\zeta^2+\zeta^3$	1
16	$-\zeta x+2y-1$	$g_1^2g_2^4g_4g_6^3g_9^6g_{15}^5g_{17}^6$
17	$-\zeta^3x+2y-1$	1
18	$(1-\zeta^3)x+2y-\zeta^2-\zeta^3$	$g_2g_5^3$
19	$2y+1+\zeta^2-\zeta^3$	$g_1^6g_2^4g_3^6g_4^5$
20	$2y+1+\zeta-\zeta^2$	1

i	H_i	c_i
21	$(-1-\zeta^2)x+2y-1+\zeta^2$	$g_1^3g_2^4g_3^6g_6^3$
22	$(-\zeta-\zeta^3)x+2y$	$g_3^6g_5^3g_6^6g_7^6g_8^4g_9^4g_{10}^4g_{14}g_{15}^5g_{16}^3g_{17}^7g_{18}^5g_{19}^7g_{20}g_{21}^3g_{22}^3g_{23}^6g_{24}^7g_{25}^6$
23	$(\zeta+\zeta^3)x+2y+2+\zeta+\zeta^3$	$g_1^5g_2^5g_3g_4^3g_5^7g_9g_{10}^6g_{11}^6g_{14}^4g_{15}^4g_{16}^4g_{17}^4g_{18}^7g_{19}^3g_{20}^3g_{21}^7g_{22}g_{23}^2g_{24}^5g_{25}^2$
24	$(1+\zeta^2)x+2y-1+\zeta^2$	$g_1^2g_2g_3^4g_4^3g_5g_8^7g_9^5g_{10}^4g_{11}^4g_{14}^5g_{15}^6g_{16}^2g_{17}^5g_{18}^7g_{19}^2g_{20}^6g_{21}^5g_{22}^5g_{25}^7$
25	$(-\zeta+\zeta^3)x+2y$	$g_1^2g_2^7g_3^4g_4g_5g_{22}^5g_{23}^5$
26	y	$g_1^4g_2^5g_3^4g_4^6g_5^4g_6^5g_7^5g_8^2g_9^7g_{13}^4g_{14}^5g_{15}^3g_{17}^7g_{18}^7g_{20}^2g_{21}^7g_{22}^6g_{24}^3g_{25}^5$
27	$2y-1$	$g_1^6g_3^3g_5^5g_8^2g_9^2g_{10}^2g_{11}^4g_{14}g_{15}^4g_{17}^7g_{18}^6g_{19}^3g_{20}^2g_{21}^3g_{22}^6g_{23}^3g_{24}^6g_{25}^7$
28	$(-\zeta+\zeta^3)x+2y+\zeta-\zeta^3$	$g_1^6g_2^3g_3^7g_8^3g_{14}^3g_{16}g_{22}^3g_{23}^3$
29	$x+1+\zeta-\zeta^3$	$g_2^4g_3^7g_4^4g_5^2g_8^5g_{10}^3g_{11}^3g_{14}^5g_{15}^5g_{16}^3g_{17}^4g_{18}^3g_{19}^3g_{20}^2g_{21}^2g_{22}^5g_{23}^5g_{24}^6g_{25}^6$
30	$x+1$	$g_2^5g_3g_4g_5^5g_6^4g_7^5g_8^6g_9^2g_{10}^5g_{11}^5g_{15}^3g_{16}^5g_{17}^4g_{18}^5g_{19}^2g_{20}^7g_{22}^7g_{23}^3g_{24}^4$
31	$x+\zeta-\zeta^2+\zeta^3$	$g_1^5g_2^7g_3^6g_4^4g_{17}^5$
32	$x-\zeta+\zeta^2-\zeta^3$	1
33	$x+\zeta+\zeta^2-\zeta^3$	$g_1^2g_2^7g_3^6g_4^2g_7^7g_{17}^7$
34	$x-\zeta+\zeta^2+\zeta^3$	1
35	$x+\zeta$	$g_1^2g_2^7g_3^7g_5^2g_6^6g_{15}^5g_{17}^4$
36	$x-\zeta$	1
37	$x+\zeta-\zeta^2$	$g_1^5g_2^6g_3^6g_6^3g_{14}^7g_{16}^7$
38	$x-\zeta-\zeta^2$	1
39	$x+\zeta^2$	$g_1g_2^4g_3^3g_4^7g_5^5g_7^4g_{10}^2g_{11}^6g_{14}^5g_{17}^7g_{23}^5g_{24}^7$
40	x	$g_1g_2^4g_3^2g_4^7g_5^6g_6^7g_7^7g_{14}^2g_{17}^6g_{22}^4g_{24}^4$
41	$-\zeta^3x+2y-\zeta^3$	$g_1^4g_2^2g_3^6g_5^2$
42	$\zeta x+2y+\zeta$	1
43	$(\zeta+\zeta^3)x+2y-\zeta^2+\zeta^3$	$g_1^4g_2g_3g_5^6g_9^7$
44	$(\zeta+\zeta^3)x+2y+\zeta+\zeta^2$	1
45	$(1+\zeta)x+2y-1+\zeta$	$g_1^7g_2^7g_3^3g_{16}^4$
46	$(1-\zeta^3)x+2y-1-\zeta^3$	1
47	$(\zeta^2-\zeta^3)x+2y-\zeta-\zeta^3$	$g_1^5g_2^3g_4^5g_5^7$
48	$(-\zeta-\zeta^2)x+2y-\zeta-\zeta^3$	1
49	$x+2y-1$	$g_1^2g_2^2g_4^6g_5^5g_6^2g_7^6g_9^5g_{14}^5g_{15}^3g_{16}^4g_{17}^6g_{22}^5g_{24}$
50	$(1+\zeta^2)x+2y+\zeta+\zeta^3$	$g_1^6g_2^7g_3^4g_5^6g_6^2g_9^7$
51	$(1-\zeta^2)x+2y+\zeta+\zeta^3$	1
52	$(1-\zeta^3)x+2y-1-\zeta$	$g_1^5g_2^7g_3^4g_{16}^4g_{17}^4$
53	$(1+\zeta)x+2y-1+\zeta^3$	1
54	$-\zeta x+2y-\zeta^2$	$g_1^2g_2^6g_3^7g_4^4g_{16}$
55	$-\zeta^3x+2y+\zeta^2$	1
56	$\zeta^2x+2y+\zeta$	$g_1^2g_2^6g_3^5g_7^2g_8^5$
57	$-\zeta^2x+2y-\zeta^3$	1
58	$(-1+\zeta+\zeta^2-\zeta^3)x+2y-1+\zeta-\zeta^2-\zeta^3$	$g_1^3g_2^6g_3^6g_9^7g_{15}^4g_{16}^5$
59	$(-1-\zeta+\zeta^2+\zeta^3)x+2y-1-\zeta-\zeta^2+\zeta^3$	1
60	$(-1-\zeta)x+2y+\zeta+\zeta^2$	$g_2^2g_3^7g_4^5g_6^2g_{15}^5g_{16}^6$
61	$(-1-\zeta^3)x+2y-\zeta^2+\zeta^3$	1
62	$2y+\zeta+\zeta^2$	$g_2g_3^7g_4^5$
63	$2y-\zeta^2+\zeta^3$	1
64	$2x-1-\zeta-\zeta^2+\zeta^3$	$g_1^2g_3g_4^5g_5^6g_{14}^3g_{15}^4$
65	$2x-1+\zeta-\zeta^2-\zeta^3$	1
66	$2x+1-\zeta+\zeta^2-\zeta^3$	$g_1^5g_2^4g_3^5g_5^7g_{14}^5g_{17}$
67	$2x+1+\zeta+\zeta^2+\zeta^3$	1
68	$2x-1-\zeta+\zeta^2-\zeta^3$	$g_1^3g_2^2g_3^6g_{17}^4$
69	$2x-1+\zeta+\zeta^2+\zeta^3$	1
70	$x-1$	$g_{14}^6g_{17}^2$
71	$x+1-\zeta^2$	$g_3^4g_4^6g_5g_7^4g_{10}^2g_{11}^6g_{14}g_{17}^3$
72	$2y+1+\zeta^2$	$g_2^7g_7$
73	$2x+1+\zeta^2$	$g_1^7g_2g_3^6g_4^6g_5^2g_7^2g_{10}^4g_{11}^4g_{14}^5g_{15}^4g_{16}^2g_{22}^3g_{25}$
74	$(1-2\zeta^2)x+2y-1$	$g_1^7g_2g_5^3g_7^3g_{10}^4g_{11}^4g_{17}^3$
75	$-\zeta^3x+y+\zeta^2$	$g_1^3g_2^2g_3^4g_6g_8^7g_{17}^7$
76	$2x-1+\zeta^2$	$g_1^6g_2^2g_4^3g_5g_7^7g_{10}g_{11}^3g_{15}^4g_{16}^4g_{19}^4g_{20}^4g_{22}^4g_{25}^4$

The 519 edges: the row of i lists the lines $j>i$ joined to i, with the weight e_{ij} as exponent; the

edge contributes $e_{ij}(H_i, H_j)_8$ to B.
0: $3^3\ 17^1\ 22^4\ 28^6\ 29^7\ 31^5\ 32^2\ 40^3\ 48^4\ 63^6\ 65^7\ 67^7\ 68^7\ 69^1\ 71^1$ **1**: $4^6\ 8^4\ 25^2\ 32^3\ 33^7\ 35^4\ 59^5\ 64^1\ 73^6$ **2**: $23^5\ 33^3\ 39^3\ 42^3\ 67^2$ **3**: $21^6\ 22^3\ 29^5\ 30^5\ 37^3\ 45^3\ 65^5\ 66^5$ **4**: $10^5\ 22^3\ 26^6\ 32^2\ 35^3\ 73^1$ **5**: $8^3\ 12^1\ 15^1\ 23^5\ 30^2\ 32^6\ 35^5\ 41^1\ 45^7\ 64^7$ **6**: $7^3\ 11^5\ 22^5\ 34^5\ 36^4\ 38^4\ 39^5\ 48^5\ 54^5\ 63^7\ 64^1\ 67^3\ 69^6\ 74^3$ **7**: $11^3\ 17^4\ 18^3\ 20^5\ 28^7\ 30^1\ 31^3\ 33^6\ 38^1\ 40^4\ 42^6\ 43^7\ 55^2\ 62^3\ 63^5\ 66^1\ 67^4\ 68^7\ 70^2\ 71^6\ 76^5$ **8**: $22^6\ 33^3\ 39^2\ 50^4\ 61^7\ 63^2\ 64^3\ 65^3\ 73^7$ **9**: $10^6\ 24^1\ 25^3\ 27^3\ 29^7\ 34^7\ 36^3\ 37^2\ 41^1\ 44^6\ 45^6\ 46^7\ 50^3\ 56^4\ 61^1\ 63^5\ 64^2\ 68^1\ 70^2$ **10**: $24^1\ 42^3\ 45^6\ 63^4\ 66^6\ 71^2\ 74^2$ **11**: $19^5\ 20^5\ 29^5\ 31^3\ 36^5\ 39^1\ 43^7\ 54^1\ 56^4\ 63^3\ 65^7\ 66^1\ 68^3\ 73^5$ **12**: $13^2\ 22^5\ 23^7\ 24^2\ 28^3\ 34^7\ 38^4\ 39^7\ 47^1\ 58^1\ 63^6\ 74^6$ **13**: $23^2\ 29^7\ 31^6\ 34^5\ 66^7\ 72^7\ 74^2$ **14**: $15^6\ 27^2\ 28^3\ 32^6\ 34^5\ 40^3\ 46^3\ 65^5\ 72^7\ 74^7$ **15**: $18^3\ 24^2\ 34^5\ 36^1\ 39^2\ 43^2\ 45^7\ 57^1\ 59^1\ 63^3\ 76^2$ **16**: $21^4\ 22^5\ 25^7\ 28^7\ 29^2\ 31^4\ 33^7\ 34^1\ 35^5\ 37^4\ 39^7\ 48^2\ 50^6\ 55^5\ 57^4\ 59^2\ 60^3\ 65^7\ 67^6\ 68^2\ 73^6\ 76^4$ **17**: $23^6\ 29^2\ 31^3\ 33^1\ 53^2\ 56^2\ 59^2\ 65^1\ 66^6\ 69^1\ 71^6$ **18**: $22^6\ 24^3\ 29^3\ 30^5\ 37^5\ 40^5\ 66^6\ 67^5\ 76^3$ **19**: $36^3\ 67^3\ 76^3$ **20**: $29^3\ 31^5\ 34^3\ 35^3\ 38^3\ 50^5\ 59^5\ 65^3$ **21**: $22^7\ 23^5\ 29^4\ 35^1\ 39^2\ 42^7\ 44^1\ 49^3\ 62^5\ 67^6\ 70^1\ 72^2$ **22**: $24^5\ 25^3\ 28^2\ 33^4\ 36^6\ 39^2\ 52^7\ 53^2\ 56^1\ 59^7\ 64^4\ 65^5\ 67^4\ 68^1\ 72^4\ 73^1\ 76^1$ **23**: $29^1\ 32^6\ 33^3\ 35^2\ 48^7\ 50^2\ 56^5\ 66^2\ 72^5\ 74^3$ **24**: $25^2\ 35^7\ 39^3\ 42^4\ 49^2\ 51^2\ 52^4\ 53^3\ 58^6\ 62^7\ 73^6\ 76^6$ **25**: $35^4\ 38^2\ 39^7\ 50^5\ 54^5\ 63^3\ 65^3\ 66^2\ 68^2\ 72^6\ 76^1$ **26**: $33^6\ 41^3\ 47^7\ 64^4\ 67^1\ 69^5\ 73^1\ 74^1\ 76^7$ **27**: $33^2\ 35^1\ 41^4\ 43^6\ 45^5\ 48^5\ 52^4\ 57^4\ 61^7\ 67^4\ 69^1\ 71^1$ **28**: $31^6\ 34^2\ 35^3\ 39^5\ 44^5\ 62^7\ 64^1\ 67^2\ 69^2$ **29**: $41^6\ 47^1\ 48^3\ 50^2\ 51^5\ 55^3\ 58^2\ 59^3\ 72^1\ 74^4$ **30**: $51^2\ 60^6\ 74^3$ **31**: $45^7\ 49^5\ 52^6\ 61^3\ 62^6\ 75^1$ **32**: $43^1\ 48^3\ 50^5\ 55^2\ 58^3\ 61^4\ 63^4$ **33**: $41^6\ 44^1\ 46^3\ 49^1\ 50^5\ 53^1\ 56^7\ 59^1\ 61^5\ 62^7$ **34**: $41^3\ 45^2\ 47^7\ 50^5\ 57^3\ 61^7$ **35**: $41^5\ 42^4\ 43^7\ 44^5\ 45^2\ 48^6\ 49^3\ 51^3\ 52^4\ 60^2\ 62^1\ 63^5$ **36**: $41^5\ 43^1\ 45^3\ 47^6\ 57^2\ 63^3$ **37**: $41^4\ 42^5\ 46^5\ 49^6\ 56^1\ 60^1\ 74^4$ **38**: $46^7\ 51^5\ 54^1\ 57^3\ 63^5\ 75^1$ **39**: $43^1\ 49^5\ 50^2\ 53^5\ 55^3\ 74^6\ 75^1$ **40**: $44^1\ 51^5\ 60^4\ 62^5\ 72^7\ 75^1$ **41**: $45^2\ 48^2\ 54^6\ 58^6\ 60^3\ 71^7\ 73^5$ **42**: $43^6\ 60^1\ 62^1\ 66^2\ 71^1\ 72^4\ 74^2\ 76^1$ **43**: $47^2\ 51^2\ 58^2\ 73^2\ 76^5$ **44**: 53^3 **45**: $51^7\ 52^3\ 59^1\ 60^2\ 61^7\ 66^6\ 67^7\ 69^7\ 70^6\ 71^6\ 73^2\ 74^7$ **46**: $48^2\ 55^1\ 62^1$ **47**: $57^4\ 68^7\ 71^6\ 76^2$ **48**: $52^3\ 61^3\ 65^7\ 66^5\ 69^7\ 71^3\ 73^7\ 74^4\ 76^6$ **49**: $69^5\ 71^6\ 72^3\ 76^7$ **50**: $65^5\ 66^6\ 69^6$ **51**: $63^5\ 66^2\ 68^4\ 69^6\ 74^4$ **52**: $53^5\ 61^6\ 62^6\ 65^7\ 68^5\ 69^6$ **53**: $61^6\ 63^4\ 69^2\ 73^3$ **54**: $56^7\ 60^2\ 64^7\ 65^6\ 66^5\ 71^4\ 74^4\ 76^1$ **55**: 68^3 **56**: $62^7\ 76^5$ **57**: $65^5\ 71^1\ 76^6$ **59**: $67^1\ 72^3$ **60**: $70^5\ 74^6\ 76^6$ **61**: $73^4\ 76^1$ **62**: $65^2\ 66^4\ 67^6\ 68^3\ 76^3$ **63**: $64^7\ 71^6\ 73^6$ **64**: 72^3 **65**: 75^7 **67**: 75^7 **69**: 72^3 **70**: $72^3\ 74^1\ 75^6$ **71**: 74^2 **72**: $74^5\ 76^3$ **74**: 76^1

Table 4: for each orbit, the representative ℓ, its stabilizer Γ, the parameter t, the special points removed from f_ℓ (with multiplicity if it exceeds 1), $\deg \psi_\ell$, and W_ℓ with $P_\ell \equiv W_\ell^8 \pmod{\psi_\ell}$.

1. $\sigma_5(H_0)$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg \psi = 3$
$W = 133088640 - 215061424\zeta + 168590176\zeta^2 - 24519824\zeta^3 + (180441680 - 372076768\zeta + 342363120\zeta^2 - 118693984\zeta^3)t + (415558768 - 165525264\zeta - 178190288\zeta^2 + 423430032\zeta^3)t^2$

2. H_2, $\Gamma = \{1\}$, $t = x$, removed: R_5, $\deg \psi = 3$
$W = -583140 - 470202\zeta - 81804\zeta^2 + 354534\zeta^3 + (-946830 - 230160\zeta + 621330\zeta^2 + 1108908\zeta^3)t + (107022 + 465942\zeta + 551862\zeta^2 + 314514\zeta^3)t^2$

3. $\sigma_3(H_4)$, $\Gamma = \{1\}$, $t = x$, removed: Q_0, $\deg \psi = 3$
$W = \frac{1}{128}\big(-5134107 + 3055421\zeta + 920273\zeta^2 - 4265787\zeta^3 + (-7249848 + 5761206\zeta - 1023964\zeta^2 - 4355882\zeta^3)t + (-3289977 + 3064901\zeta - 1007713\zeta^2 - 1645587\zeta^3)t^2\big)$

4. $\sigma_3(H_6)$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg \psi = 3$
$W = \frac{1}{8192}\big(-5186728998 - 113443251712\zeta + 164635742530\zeta^2 - 120004279022\zeta^3 + (-36766402378 - 51568400947\zeta + 107214252570\zeta^2 - 102219430225\zeta^3)t + (222034749450 - 141733785873\zeta - 22675477662\zeta^2 + 172354031637\zeta^3)t^2\big)$

5. $\sigma_3(H_8)$, $\Gamma = \{1\}$, $t = x$, removed: R_3, $\deg \psi = 3$
$W = 81733131108 - 31823694288\zeta - 50510513196\zeta^2 + 76949088912\zeta^3 + (-77661373320 + 90652263156\zeta - 29147732496\zeta^2 - 60284599596\zeta^3)t + (108290848428 - 115764568308\zeta + 83786513508\zeta^2 - 4260927780\zeta^3)t^2$

6. $\sigma_5(H_{10})$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg \psi = 3$
$W = \frac{1}{128}\big(853728 + 8658452\zeta + 7105342\zeta^2 + 3643234\zeta^3 + (-3530089 - 909979\zeta - 4980999\zeta^2 + 4450335\zeta^3)t + (13435279 + 7631325\zeta - 193335\zeta^2 - 6344689\zeta^3)t^2\big)$

7. H_{12}, $\Gamma = \{1\}$, $t = x$, removed: R_7, $\deg \psi = 3$
$W = -424410672 + 205506208\zeta + 710681032\zeta^2 + 802799672\zeta^3 + (1512178024 + 912539688\zeta - 224778044\zeta^2 - 1222325852\zeta^3)t + (-707985920 - 1066787728\zeta - 801539148\zeta^2 - 62929492\zeta^3)t^2$

8. $\sigma_5(H_{14})$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg \psi = 3$
$W = -1612699800240 - 39595423848\zeta + 1556850443328\zeta^2 + 2241030782136\zeta^3 + (-269515174600 - 3238030089792\zeta - 4310814099016\zeta^2 - 2857850498272\zeta^3)t + (1869374497448 + 1972745847352\zeta + 919946284152\zeta^2 - 671003391080\zeta^3)t^2$

9. $\sigma_5(H_{16})$, $\Gamma = \{1\}$, $t = x$, removed: Q_0, R_7, $\deg \psi = 2$
$W = \frac{1}{2048}\big(-2191915547005 - 4134417396375\zeta - 7066775616535\zeta^2 - 6384353527250\zeta^3 + (4413128067205 + 9016816659168\zeta + 9003709130935\zeta^2 + 823362383201\zeta^3)t\big)$

10. H_{18}, $\Gamma = \{1\}$, $t = x$, removed: R_3, $\deg \psi = 3$
$W =$
$15 - 1336\zeta - 1904\zeta^2 - 1361\zeta^3 + (5772 - 424\zeta - 6383\zeta^2 - 8597\zeta^3)t + (1701 - 1196\zeta - 3397\zeta^2 - 3610\zeta^3)t^2$

11. H_{19}, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = \frac{1}{2}\big(-108 - 114\zeta - 42\zeta^2 + 24\zeta^3 + (-135 + 171\zeta + 387\zeta^2 + 363\zeta^3)t + (-444 - 294\zeta + 24\zeta^2 + 330\zeta^3)t^2 + (132 - 78\zeta - 228\zeta^2 - 258\zeta^3)t^3\big)$

12. H_{21}, $\Gamma = \{1,5\}$, $t = x$, $\deg\psi = 4$
$W = -2590032 + 3219912\zeta - 1951824\zeta^2 - 467016\zeta^3 + (7415272 - 1672792\zeta - 5001496\zeta^2 + 8808856\zeta^3)t + (4053936 - 9243416\zeta + 8976352\zeta^2 - 3448712\zeta^3)t^2 + (-5225032 + 3908072\zeta - 304824\zeta^2 - 3493896\zeta^3)t^3$

13. H_{22}, $\Gamma = \{1,3\}$, $t = x$, $\deg\psi = 4$
$W = 92914989515140260702080000 - 18874049118239605446912000\zeta + 13013815517437706460608000\zeta^2 + 75524052352582367766400000\zeta^3 + (-24880510949735356456640000 + 33500451056447344941440000\zeta + 24773498118044366364800\zeta^2 - 31758480936152532280320000\zeta^3)t + (-53306371893111077814080000 + 20281076107373970291200\zeta + 15857006655488458144160000\zeta^2 - 11563868813374988906880000\zeta^3)t^2 + (16856798762689425451968000 - 12488438656399578332096000\zeta - 24117553789766758741504000\zeta^2 + 28491704762844827682752000\zeta^3)t^3$

14. H_{23}, $\Gamma = \{1,3\}$, $t = x$, $\deg\psi = 4$
$W = \frac{1}{3529}\big(-49605165924550579226441 25 - 47878703837996323621065 66\zeta - 14345018031035519622346 86\zeta^2 - 14530615569591471509620 23\zeta^3 + (58502311271108270942 1912 - 84385218113536472590238 55\zeta - 47204971320609125672674 41\zeta^2 - 76973917213469629703071 26\zeta^3)t + (51283896133681697872221 99 - 28419581408269659009868 62\zeta - 26106682449455339673415 32\zeta^2 - 53058493081764548162333 01\zeta^3)t^2 + (22047850152724435919773 74 - 45137662411434484982 4981\zeta - 67031581869481855133207 7\zeta^2 - 15187787866773272510893 98\zeta^3)t^3\big)$

15. $\sigma_3(H_{24})$, $\Gamma = \{1,5\}$, $t = x$, $\deg\psi = 4$
$W = 26147633656428774638509126117553986407860160 + 34765092014512603182501904258958252809574 40\zeta - 93967150531462956687563377281395921097475 2\zeta^2 - 30586837814328636323701174098775642471574112\zeta^3 + (22530060921613782276478184479242492795174432 + 58178002742731221294228192342035733991926528\zeta + 46616133531464983757727738737327148759917280\zeta^2 - 18065958587989466091312672746984599149697920\zeta^3)t + (-39891689407982103952901552193026676755333664 + 23770043668912044067284627450976480759374048\zeta + 26419383721376698592846706461381245676357376\zeta^2 + 51035812926892285527202566984494115457855296\zeta^3)t^2 + (-22927211634018797596895289407269149062662848 - 16773907624201631837420809275453137965404000\zeta - 10972549695578091693282316518010896052590912\zeta^2 + 24760568406158776760613594664638052621151136\zeta^3)t^3$

16. $\sigma_3(H_{25})$, $\Gamma = \{1,7\}$, $t = x$, $\deg\psi = 4$
$W = 1652648581135104 - 93684635260416\zeta - 1520158499085312\zeta^2 + 2243513401585920\zeta^3 + (-272733046138368 + 14482361056512\zeta + 252251894488320\zeta^2 - 371220411717120\zeta^3)t + (-1316125480714752 + 74781684037632\zeta + 1210368209329152\zeta^2 - 1786500820835328\zeta^3)t^2 + (-3052168954104576 + 174876783763200\zeta + 28048555834776320\zeta^2 - 4141541946262272\zeta^3)t^3$

17. H_{26}, $\Gamma = \{1,3,5,7\}$, $t = x$, removed: R_1, R_3, R_5, R_7, $\deg\psi = 0$
$\psi = 1$

18. H_{27}, $\Gamma = \{1,3,5,7\}$, $t = x$, removed: Q_0, $\deg\psi = 3$
$W = 22034151039020459295483778207768368181260384466362646732 5 + 15919212065139890470721539871963100825174681651850577274 0\zeta - 29117432428304255372274151875977494096362748126409926185\zeta^2 + 14496504536420434126727612241696219848999546833052363920\zeta^3 + (28227065148333236616797656749644034671556210657981485763 0 + 19261487212544504322886385272179615978663696136590218106 0\zeta - 89970640339854140225051896562049650286772795214363545 20\zeta^2 + 15910752932857721205948368440284085907976974178669839650\zeta^3)t + (32403218396944380218726005379525681173162233939727284351 0 + 26360756306308050710986133820389912257350434045466055251 0\zeta - 39878396800617949356857710997087106177182640427018850850\zeta^2 + 19715674759219691389330681571817467787933103150409109627 0\zeta^3)t^2$

19. $\sigma_3(H_{28})$, $\Gamma = \{1,7\}$, $t = x$, $\deg\psi = 4$
$W = -1237997384236320 + 1737780736962880\zeta - 1219600219000768\zeta^2 - 12984845532704\zeta^3 + (2457294767261712 - 2377700486575440\zeta + 905300072671952\zeta^2 + 1097327676223312\zeta^3)t + (-2095535796638240 + 1372889921357568\zeta + 153955944540544\zeta^2 - 1590523349158304\zeta^3)t^2 + (612318030212784 + 114368648107024\zeta - 774047393622480\zeta^2 + 980243107659056\zeta^3)t^3$

20. H_{29}, $\Gamma = \{1,7\}$, $t = y$, $\deg\psi = 3$
$W = 21015112091080153232270341191278045401894656 + 11368357943381249431230214766101006892888832\zeta - 49378261056384795803625174468038996165268 48\zeta^2 - 18351498590615113361187021983389443222382848\zeta^3 +$

$(-161589992421080642489201815775732165445043 2 + 2900180761274297019089813850206678524959360\zeta + 571737489013843652726435504180733866655436 8\zeta^2 + 518540834953086602967474990767562432776025 6\zeta^3)t + (-411166241102797189280717887372964331632256 0 - 418404800525185817383547250029275584216652 8\zeta - 180547502361930427889287217119968093303014 4\zeta^2 + 163072074032356438024759134382324512962304 0\zeta^3)t^2$

21. H_{30}, $\Gamma = \{1,3,5,7\}$, $t = y$, $\deg\psi = 3$
$W = 54452718558650164914084511807562095845279909590251845000000000 + 960617957123391260634586701299402590542673781314871775000000000\zeta + 263431346482609401488448343441379391747969097171458150000000000\zeta^2 + 276112046692297160160542407040156154440134649711269155000000000\zeta^3 + (-60627610761915994616612572102806090917156503004999727000000000 - 41038858955140288937930048255512182802527997936173046650000000 0\zeta + 59460218206074693478821197519625634137398739075843411000000000\zeta^2 - 39894516252118981315971906720630927257509949387940030000000 00\zeta^3)t + (-15268265796134912249534557048439132093147596652386554000000000 + 34771220480794227925920777647034113441282355887594047200000000 0\zeta - 11701574329472410615387752071932525984466638963569493800000000 0\zeta^2 - 45206786047795877249098692194483361255483642648544676100000000 0\zeta^3)t^2$

22. $\sigma_3(H_{31})$, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = 76321440 + 30957120\zeta - 32360256\zeta^2 - 76617792\zeta^3 + (-24771408 + 1864464\zeta + 27493488\zeta^2 + 36501936\zeta^3)t + (-7706816 - 14008928\zeta - 13085056\zeta^2 - 3678336\zeta^3)t^2$

23. H_{33}, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = -63626432 - 130072480\zeta + 32245824\zeta^2 + 97403360\zeta^3 + (198483104 + 60214464\zeta - 102099328\zeta^2 - 106486048\zeta^3)t + (-176468608 - 158307936\zeta - 149758144\zeta^2 + 45525152\zeta^3)t^2$

24. H_{35}, $\Gamma = \{1\}$, $t = y$, removed: R_5, $\deg\psi = 2$
$W = 15544 - 26008\zeta - 27772\zeta^2 - 46716\zeta^3 + (83304 + 93032\zeta + 78448\zeta^2 + 35344\zeta^3)t$

25. H_{37}, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = 34949 - 10325\zeta - 24407\zeta^2 + 43275\zeta^3 + (-6841 - 4693\zeta + 19563\zeta^2 - 15751\zeta^3)t + (-2276 + 15624\zeta - 16532\zeta^2 + 3668\zeta^3)t^2$

26. H_{39}, $\Gamma = \{1,5\}$, $t = y$, $\deg\psi = 3$
$W = 10774878767640662400 - 1899453562861852 80\zeta - 10190225009281996800\zeta^2 + 14380339661456376960\zeta^3 + (4040648872073573760 - 20538093406098422880\zeta + 25554965540088631680\zeta^2 - 14440089849483687840\zeta^3)t + (-26274057725431866240 + 23854379397496718400\zeta - 87102786722402563 20\zeta^2 - 12420703577475388800\zeta^3)t^2$

27. H_{40}, $\Gamma = \{1,3,5,7\}$, $t = y$, removed: Q_0, $\deg\psi = 2$
$W = 18051161622878427634368896484375 - 7380972926897248997314453125000\zeta - 25649009594973101151397705078125\zeta^2 + 31615910370422772372729492187500\zeta^3 + (40658788596384895215069580078125 - 46442829313389263271240234375000\zeta + 11145911059929481960308837890625\zeta^2 + 83057149662763173699462890625 00\zeta^3)t$

28. $\sigma_3(H_{41})$, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = \frac{1}{4096}\big(-80242789621 - 1802001209\zeta + 82780572607\zeta^2 - 115285602165\zeta^3 + (-217334824301 + 184954724427\zeta - 44268813140\zeta^2 - 122380580518\zeta^3)t + (-204459379635 + 261596550973\zeta - 165533651017\zeta^2 - 27521026141\zeta^3)t^2 + (-71856306035 + 168834338653\zeta - 166930158234\zeta^2 + 67238393884\zeta^3)t^3\big)$

29. $\sigma_5(H_{43})$, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = 1740162 + 1264029\zeta + 98901\zeta^2 - 1028676\zeta^3 + (6091250 + 3282403\zeta - 1505103\zeta^2 - 5758526\zeta^3)t + (5667147 + 1716633\zeta - 2262591\zeta^2 - 5539257\zeta^3)t^2 + (2781834 + 1622990\zeta - 876442\zeta^2 - 2118884\zeta^3)t^3$

30. $\sigma_3(H_{45})$, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = 3562807 + 9561407\zeta - 17153613\zeta^2 + 14584015\zeta^3 + (4689017 - 9978785\zeta + 9255100\zeta^2 - 3191582\zeta^3)t + (-44405975 + 16005181\zeta + 21780687\zeta^2 - 46711823\zeta^3)t^2 + (-9557665 + 28112437\zeta - 30134594\zeta^2 + 14561282\zeta^3)t^3$

31. $\sigma_5(H_{47})$, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = \frac{1}{4}\big(-8465391 + 16210044\zeta - 14230323\zeta^2 + 5163264\zeta^3 + (22511520 - 33560730\zeta + 28431243\zeta^2 - 1780461\zeta^3)t + (-38570904 + 21421179\zeta + 10527489\zeta^2 - 31010688\zeta^3)t^2 + (2728404 + 15882480\zeta - 24200370\zeta^2 + 19429308\zeta^3)t^3\big)$

32. H_{49}, $\Gamma = \{1,3,5,7\}$, $t = x$, removed: Q_0, $\deg\psi = 3$
$W = \frac{1}{2048}\big(5325542058824523741974386380160535156 25 + 56575926503616907740127812868910460937 50\zeta - 38973624022828349061859662464036953125 0\zeta^2 + 69469093968603542888369788656876796875 0\zeta^3 + (-69368674988379170323884362257144687500 0 - 71420749996320721144871392070468812500 00\zeta + 50617611509234295498532927030479375000 0\zeta^2 - 91479544179883952951704142439410625000 0\zeta^3)t +$

$(60666471167822702006411295820314492187 5 + 619990351688838556177392851657404453125 0\zeta - 495740981060471937762885550144577343750\zeta^2 + 80047747693281047403099534245733515625 0\zeta^3)t^2\big)$

33. $\sigma_5(H_{50})$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W = 313032536336 - 18426373808\zeta - 287317050048\zeta^2 + 424818320944\zeta^3 + (-819456405680 + 697514807568\zeta - 165562476560\zeta^2 - 462767549024\zeta^3)t + (-822253110368 + 868257463776\zeta - 404909004176\zeta^2 - 294404969168\zeta^3)t^2$

34. H_{52}, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W = 155262172 - 26526432\zeta - 192776252\zeta^2 - 246100416\zeta^3 + (83628472 + 354685508\zeta + 417972536\zeta^2 + 236416956\zeta^3)t + (-91637076 - 255344996\zeta - 269475116\zeta^2 - 125750492\zeta^3)t^2$

35. $\sigma_5(H_{54})$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W = \frac{1}{8}\big(-47066 + 58382\zeta - 35146\zeta^2 - 8322\zeta^3 + (77147 - 411509\zeta + 505033\zeta^2 - 302871\zeta^3)t + (-480177 + 150059\zeta + 268579\zeta^2 - 529589\zeta^3)t^2\big)$

36. $\sigma_3(H_{56})$, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W = \frac{1}{32}\big(-1038380 + 1240372\zeta - 723310\zeta^2 - 224616\zeta^3 + (1183537 - 1928246\zeta + 1550879\zeta^2 - 261372\zeta^3)t + (-748777 + 1107686\zeta - 818519\zeta^2 + 58872\zeta^3)t^2\big)$

37. H_{58}, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W =$
$-60310237254 - 45495018918\zeta + 124649908422\zeta^2 - 130786576266\zeta^3 + (91920318110 - 255898904258\zeta + 269975367890\zeta^2 - 125903941574\zeta^3)t + (91408387200 - 131437556344\zeta + 94472385512\zeta^2 - 2166582776\zeta^3)t^2$

38. H_{60}, $\Gamma = \{1\}$, $t = x$, removed: R_1, $\deg\psi = 3$
$W = \frac{1}{2}\big(-5411152568 - 9334906197\zeta - 7790402591\zeta^2 - 1682401718\zeta^3 + (38304294596 + 41623967781\zeta + 20560810583\zeta^2 - 12546538900\zeta^3)t + (-15292992736 - 18040318040\zeta - 10219883064\zeta^2 + 3587261710\zeta^3)t^2\big)$

39. $\sigma_5(H_{62})$, $\Gamma = \{1\}$, $t = x$, $\deg\psi = 4$
$W = 5633904 - 5484764\zeta + 2123768\zeta^2 + 2482892\zeta^3 + (771412 + 4223384\zeta - 6746356\zeta^2 + 5315880\zeta^3)t + (-4734988 + 3968588\zeta - 876156\zeta^2 - 2729764\zeta^3)t^2 + (-1506688 - 1885488\zeta + 4173992\zeta^2 - 4015208\zeta^3)t^3$

40. H_{64}, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = \frac{1}{8}\big(39771 - 247770\zeta - 373041\zeta^2 - 310158\zeta^3 + (119592 + 7380\zeta - 86292\zeta^2 - 100944\zeta^3)t + (505188 + 764172\zeta + 476712\zeta^2 - 24444\zeta^3)t^2\big)$

41. $\sigma_3(H_{66})$, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = \frac{1}{8}\big(-3807643 + 327629\zeta - 2971209\zeta^2 - 7262635\zeta^3 + (75900 + 3938382\zeta - 4104024\zeta^2 + 13494534\zeta^3)t + (13946624 - 10004680\zeta + 8662224\zeta^2 - 4945000\zeta^3)t^2\big)$

42. $\sigma_3(H_{68})$, $\Gamma = \{1\}$, $t = y$, $\deg\psi = 3$
$W = \frac{1}{2}\big(-16520 - 18320\zeta - 15730\zeta^2 + 3915\zeta^3 + (-20708 - 16373\zeta + 14601\zeta^2 + 24866\zeta^3)t + (1306 + 23066\zeta + 22568\zeta^2 + 738\zeta^3)t^2\big)$

43. H_{70}, $\Gamma = \{1, 3, 5, 7\}$, $t = y$, $\deg\psi = 3$
$W = -6556832 - 199756800\zeta + 10089568\zeta^2 + 42392832\zeta^3 + (18893664 + 495174912\zeta - 22370912\zeta^2 - 41726976\zeta^3)t + (-17201280 - 252359424\zeta + 3828672\zeta^2 - 160471296\zeta^3)t^2$

44. H_{71}, $\Gamma = \{1, 5\}$, $t = y$, $\deg\psi = 3$
$W = 11596503996 - 11141776800\zeta + 11119508928\zeta^2 - 536114700\zeta^3 + (-4813310304 + 17347665456\zeta - 16857772272\zeta^2 + 9268557408\zeta^3)t + (2059203168 - 901535184\zeta + 6095803824\zeta^2 - 3860078112\zeta^3)t^2$

45. $\sigma_3(H_{72})$, $\Gamma = \{1, 5\}$, $t = x$, $\deg\psi = 4$
$W = -12240 + 60176\zeta + 95520\zeta^2 + 76432\zeta^3 + (13808 + 7616\zeta - 4144\zeta^2 - 12288\zeta^3)t + (31680 - 38608\zeta - 87840\zeta^2 - 85856\zeta^3)t^2 + (-95392 - 32512\zeta + 49456\zeta^2 + 102416\zeta^3)t^3$

46. $\sigma_3(H_{73})$, $\Gamma = \{1, 5\}$, $t = y$, $\deg\psi = 3$
$W = \frac{1}{32}\big(-762780784459585823613 - 186867824724799684854 3\zeta - 187069108169595432419 1\zeta^2 - 779204834541849196701\zeta^3 + (-268814573067402980838 0 + 991129003632374195160\zeta + 406602078908781142134 0\zeta^2 + 478299080826061975062 0\zeta^3)t + (723218359817751357463 2 + 560358234289254052396 8\zeta + 694871192936914446024\zeta^2 - 464985279637111052582 4\zeta^3)t^2\big)$

47. H_{74}, $\Gamma = \{1, 5\}$, $t = x$, removed: Q_0, $\deg\psi = 3$
$W = \frac{1}{16384}\big(-70140747198203747391 15 + 700094645884405986835 5\zeta + 245602164879436942239 30\zeta^2 - 105149541890777866406 10\zeta^3 + (-352699133646715206397 20 + 130440005604432815900 40\zeta - 237335709252424012847 10\zeta^2 + 164896181394041370764 70\zeta^3)t + (280829990119390064342 55 - 155391197870502258949 35\zeta - 153403668514466915331 60\zeta^2 + 300030862317969084192 0\zeta^3)t^2\big)$

48. $\sigma_3(H_{75})$, $\Gamma = \{1\}$, $t = x$, removed: R_5, $\deg\psi = 3$
$W = 1628 - 5630\zeta + 5674\zeta^2 - 3780\zeta^3 + (-9898 + 15938\zeta - 15354\zeta^2 + 2414\zeta^3)t + (9454 - 4688\zeta - 4688\zeta^2 + 10826\zeta^3)t^2$

49. $\sigma_3(H_{76})$, $\Gamma = \{1, 5\}$, $t = y$, $\deg\psi = 3$
$W = \frac{1}{128}\big(205882363527134071323249 + 576105373910615394946617\zeta + 510203389484382344066943\zeta^2 + 916492362357598773179 4\zeta^3 + (-425373499152766500943662 - 214231294044613870562498 4\zeta -$

$$100232363431849604163773 4\zeta^2 + 64494759301563799482912\zeta^3)t + (42817965188813302046644 8 + 17256785568777040967849 52\zeta - 13323688058630672399789 64\zeta^2 - 73189979919275390326623 6\zeta^3)t^2\big)$$

Table 5: the Frobenius checks of Proposition 13: the orbit (numbered as in Table 4), the factor q of ψ_ℓ, the group D, the Frobenius element, the prime p, the root t_0 of q modulo p and the point (x_0, y_0) modulo p. All values are 0.

orbit	factor $q(t)$	D	Frob	p	t_0	(x_0, y_0)
12	$\frac{1}{2}\big(3+3\zeta^2+(-8+6\zeta^2)t+(-7-7\zeta^2)t^2+(2-2\zeta^2)t^3+2t^4\big)$	$\{1,5\}$	σ_5	101	38	$(38,87)$
13	$\frac{1}{2}\big(2+(-\zeta-\zeta^3)t+(-\zeta-\zeta^3)t^2+(2\zeta+2\zeta^3)t^3+2t^4\big)$	$\{1,3\}$	σ_3	83	15	$(15,57)$
14	$-3+\zeta+\zeta^3+(-2-\zeta-\zeta^3)t+t^2$	$\{1,3\}$	σ_3	107	11	$(11,27)$
14	$\frac{1}{2}\big(2-3\zeta-3\zeta^3+4t+2t^2\big)$	$\{1,3\}$	σ_3	83	31	$(31,60)$
15	$\frac{1}{2}\big(3-3\zeta^2+(6+4\zeta^2)t+(-5+5\zeta^2)t^2+(-2-2\zeta^2)t^3+2t^4\big)$	$\{1,5\}$	σ_5	109	41	$(41,50)$
16	$\frac{1}{2}\big(2+(\zeta-\zeta^3)t+(\zeta-\zeta^3)t^2+(2\zeta-2\zeta^3)t^3+2t^4\big)$	$\{1,7\}$	σ_7	79	15	$(15,51)$
18	$\frac{1}{2}\big(-1+2t^3\big)$	$\{1,3,5,7\}$	σ_3	83	5	$(5,42)$
18	$\frac{1}{2}\big(-1+2t^3\big)$	$\{1,3,5,7\}$	σ_5	101	35	$(35,51)$
19	$\frac{1}{2}\big(2-3\zeta+3\zeta^3+(6\zeta-6\zeta^3)t+(-5\zeta+5\zeta^3)t^2+(2\zeta-2\zeta^3)t^3+2t^4\big)$	$\{1,7\}$	σ_7	79	2	$(2,35)$
20	$\frac{1}{4}\big(-18-12\zeta+12\zeta^3+(-\zeta+\zeta^3)t+4t^3\big)$	$\{1,7\}$	σ_7	103	36	$(64,36)$
21	$\frac{1}{2}\big(-1+2t^3\big)$	$\{1,3,5,7\}$	σ_3	83	5	$(82,5)$
21	$\frac{1}{2}\big(-1+2t^3\big)$	$\{1,3,5,7\}$	σ_5	101	35	$(100,35)$
26	$\frac{1}{4}\big(-2+(1-\zeta^2)t+4t^3\big)$	$\{1,5\}$	σ_5	149	132	$(105,132)$
27	$\frac{1}{2}\big(1+t+2t^2\big)$	$\{1,3,5,7\}$	σ_3	107	24	$(0,24)$
27	$\frac{1}{2}\big(1+t+2t^2\big)$	$\{1,3,5,7\}$	σ_5	109	69	$(0,69)$
32	$\frac{1}{2}\big(3-2t+t^2+2t^3\big)$	$\{1,3,5,7\}$	σ_3	83	21	$(21,73)$
32	$\frac{1}{2}\big(3-2t+t^2+2t^3\big)$	$\{1,3,5,7\}$	σ_5	101	24	$(24,39)$
43	$\frac{1}{2}\big(-1+t+2t^3\big)$	$\{1,3,5,7\}$	σ_3	83	20	$(1,20)$
43	$\frac{1}{2}\big(-1+t+2t^3\big)$	$\{1,3,5,7\}$	σ_5	101	19	$(1,19)$
44	$\frac{1}{4}\big(3+\zeta^2t+4t^3\big)$	$\{1,5\}$	σ_5	101	72	$(90,72)$
45	$\frac{1}{2}\big(1-3\zeta^2+(1-\zeta^2)t+2t^4\big)$	$\{1,5\}$	σ_5	101	18	$(18,45)$
46	$\frac{1}{16}\big(-3+(2+2\zeta^2)t+16t^3\big)$	$\{1,5\}$	σ_5	101	53	$(45,53)$
47	$\frac{1}{2}\big(3-8\zeta^2+(10+10\zeta^2)t+(-11+2\zeta^2)t^2+2t^3\big)$	$\{1,5\}$	σ_5	101	57	$(57,8)$
49	$\frac{1}{16}\big(-3+(6+2\zeta^2)t+16t^3\big)$	$\{1,5\}$	σ_5	101	10	$(46,10)$

Table 6: the values b_p on the smooth residue discs $(x, y) \equiv (a, b) \pmod p$ at the odd primes of S.

p	b_p	residue discs (a, b)
3	0	$(0,2)$
5	2	$(0,3)$, $(2,1)$, $(2,3)$, $(3,4)$, $(4,2)$
7	1	$(0,5)$
7	4	$(5,1)$
7	5	$(2,2)$
7	6	$(0,4)$, $(3,2)$, $(4,6)$
7	7	$(1,3)$
11	0	$(0,2)$, $(0,3)$, $(0,6)$, $(2,9)$, $(5,1)$, $(7,2)$, $(8,6)$, $(8,7)$, $(8,9)$, $(9,3)$, $(10,8)$
13	0	$(0,7)$, $(3,6)$, $(4,6)$, $(4,9)$, $(4,11)$, $(8,11)$, $(9,4)$, $(10,8)$, $(11,2)$
17	4	$(0,9)$, $(1,2)$, $(2,0)$, $(4,12)$, $(5,10)$, $(5,12)$, $(6,6)$, $(6,13)$, $(6,15)$, $(7,16)$, $(8,0)$, $(8,6)$, $(8,11)$, $(9,0)$, $(10,3)$, $(10,11)$, $(11,2)$, $(14,5)$, $(15,0)$, $(15,8)$, $(15,9)$, $(16,15)$
41	0	$(0,21)$, $(1,10)$, $(3,0)$, $(3,9)$, $(3,32)$, $(4,37)$, $(6,12)$, $(8,18)$, $(12,5)$, $(14,0)$, $(17,23)$, $(19,1)$, $(19,15)$, $(19,25)$, $(22,16)$, $(23,10)$, $(23,35)$, $(23,37)$, $(25,38)$, $(26,33)$, $(27,0)$, $(28,10)$, $(28,11)$, $(28,20)$, $(30,10)$, $(31,7)$, $(32,3)$, $(33,1)$, $(33,19)$, $(33,21)$, $(38,0)$, $(38,12)$, $(38,29)$, $(39,30)$, $(40,33)$

p	b_p	residue discs (a, b)
73	0	$(0,37)$, $(1,21)$, $(3,5)$, $(3,14)$, $(3,54)$, $(4,63)$, $(5,23)$, $(6,3)$, $(7,25)$, $(9,3)$, $(9,32)$, $(9,38)$, $(10,0)$, $(11,33)$, $(11,49)$, $(11,64)$, $(14,1)$, $(14,18)$, $(14,54)$, $(15,18)$, $(16,67)$, $(18,71)$, $(21,38)$, $(22,0)$, $(22,7)$, $(22,66)$, $(23,62)$, $(24,6)$, $(25,61)$, $(27,20)$, $(28,17)$, $(30,13)$, $(30,19)$, $(30,41)$, $(32,33)$, $(34,24)$, $(35,69)$, $(36,2)$, $(36,14)$, $(36,57)$, $(37,5)$, $(38,5)$, $(38,25)$, $(38,43)$, $(40,19)$, $(41,29)$, $(45,20)$, $(46,52)$, $(47,16)$, $(49,18)$, $(49,64)$, $(51,0)$, $(52,65)$, $(53,1)$, $(53,34)$, $(53,38)$, $(54,15)$, $(54,29)$, $(59,15)$, $(60,65)$, $(61,36)$, $(61,43)$, $(61,67)$, $(62,21)$, $(62,62)$, $(62,63)$, $(63,0)$, $(63,35)$, $(63,38)$, $(65,4)$, $(65,7)$, $(65,62)$, $(67,69)$, $(68,5)$, $(68,18)$, $(68,50)$, $(69,62)$, $(70,52)$

Table 7: the values b_2 on the 32 discs at 2: on the disc of $(0,1)$ the parameter is x, on the disc of $(1,1)$ it is y.

parameter	b_2	residues modulo 32
x	2	2, 6, 10, 14, 18, 22, 26, 30
x	6	0, 4, 8, 12, 16, 20, 24, 28
y	0	1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31

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